Estimation of factors using higher-order multi-cumulants in weak factor models

Guanglin Huang[a,d], Wanbo Lu[b,*], and Kris Boudt[c,d,e]

[a] *Joint Laboratory of Data Science and Business Intelligence, Southwestern University of Finance and Economics, Chengdu, Sichuan 611130, China*

[b] *School of Management Science and Engineering, Southwestern University of Finance and Economics, Chengdu 611130, China*

[c] *Department of Economics, Universiteit Gent, 9000 Gent, Belgium*

[d] *Department of Business, Vrije Universiteit Brussel, 1050 Brussels, Belgium*

[e] *School of Business and Economics, Vrije Universiteit Amsterdam, 1081 Amsterdam, The Netherlands*

Abstract

When factors are weak, covariance-based factor analysis methods tend to exhibit poor performance. To address this issue in the case of non-Gaussian data, we propose a new method called Higher-order multi-cumulant Factor Analysis (HFA). HFA estimates factors and factor loadings via the eigenvalue decomposition of the product of a higher-order multi-cumulant matrix and its transpose. We derive the asymptotic properties of HFA under a weak factor model where non-Gaussianity originates solely from the latent factors, while idiosyncratic errors remain Gaussian. Simulation studies demonstrate that HFA significantly improves both factor selection and estimation when factors are weak and non-Gaussian, compared with traditional methods. Applied to the FRED-MD dataset, HFA identifies factors that improve out-of-sample forecasting performance for the S&P 500 monthly equity premium.

Keywords: High-dimensional factor models; Higher-order multi-cumulants; Weak factors; Dimension reduction; Eigenanalysis; FRED-MD

1 Introduction

Factor models are widely used to characterize the low-dimensional common structure in large panels of economic data (Ludvigson and Ng, 2007; Chang et al., 2015; Giovannelli et al., 2021). Factor model approaches differ in how they estimate the latent structure. The most common approach is to use Principal Component Analysis (PCA) (Bai, 2003; Bai and Ng, 2023) and its improvements (Lettau and Pelger, 2020; Giglio et al., 2025). However, as research progressed, people recognized the drawbacks of such covariance-based factor analysis methods in weak factor models. On the one hand, as mentioned in Onatski (2012), Uematsu and Yamagata (2022) and Bai and Ng (2023), when the factors have weak explanatory power, the PCA estimators have a slow convergence rate and can even be inconsistent. This poor performance is due to the low signal-to-noise ratio of the factors when their explanatory power decreases. On the other hand, as mentioned in (Fan et al., 2022), the estimation of the number of common factors based on the difference between the eigenvalues of the covariance (or correlation) matrix becomes inconsistent when the factors' explanatory power is smaller than a threshold value, which the authors refer to as the minimal signal strength.

Given these limitations, a key question concerns the improvement of factor estimation and factor number selection in weak factor models. To this end, we propose to exploit information from the higher-order multi-cumulants to characterize the factor structure when factors are weak and drive the non-normality in the data. For such processes, we recommend using Higher-order multi-cumulant Factor Analysis (HFA), an eigenvalue-based approach designed for asymptotic settings where both cross-sectional (N) and time-series (T) dimensions diverge to infinity. By incorporating information from the non-Gaussianity of the observations, the proposed approach not only captures empirical patterns better but also enables more effective identification and inference. The asymptotic properties are derived under the restrictive assumption that any non-Gaussianity in the observed variables originates solely from the factors. In practice, we recommend using HFA instead of PCA when there is sufficient evidence that the factors are weak and that they contribute substantially to higher-order cumulants. The proposed Adaptive HFA algorithm (see Section 5.2) provides a fully data-driven procedure for choosing between PCA and HFA.

To determine the number of factors in weak factor models, we introduce Generalized Eigenvalue Ratio (GER) estimators based on adjacent eigenvalue ratios computed from the product of a higher-order multi-cumulant matrix and its transpose. The GER estimators remain consistent even when factor strengths are extremely weak, and they can detect weak factors that the covariance-based Eigenvalue Ratio (ER) estimator of Ahn and Horenstein (2013) may miss. This contrasts with mainstream factor number selection methods that rely on the covariance matrix (Fan et al., 2022; Fortin et al., 2025).

Further, we contribute to the literature on estimation and inference of weak factor models. The main approach in the literature is to focus on the PCA estimator and its improvements. Bai and Ng (2023) establish the asymptotic properties of PCA under weak factors, and Lettau and Pelger (2020) and Giglio et al. (2025) propose risk-premium PCA and supervised PCA to detect weak pricing factors. We derive the conditions under which HFA leads to consistent and asymptotically normal estimates of the factors and factor loadings. As in the linear non-Gaussian factor model of Risk et al. (2019), we assume that the non-Gaussianity of the observed variables comes only from the factors, and that the idiosyncratic errors are Gaussian. In contrast to Risk et al. (2019), we allow for the more general setting of factor dependence. We also derive the asymptotic properties of HFA for independent factor models with heterogeneous factor strengths. These statistical assumptions are consistent with structural equation models in economics and finance, where weak factors may arise from sparse loading matrices, and non-normal factors may result from nonlinear dependence among factors, heteroscedasticity, mixture distributions, structural changes, and related features.

Using the FRED-MD dataset of 124 U.S. macroeconomic and financial variables (McCracken and Ng, 2016), we demonstrate the practical utility of our approach. Empirically, this widely used dataset exhibits both weak factor strength and pronounced non-normality in the extracted factors, making it a representative setting for HFA. In simulation studies calibrated to the FRED-MD data, we find that (Adaptive) HFA recovers weak factors more accurately than PCA. In the empirical applications, HFA identifies economically meaningful factors that enhance out-of-sample forecasts of the monthly S&P 500 equity premium, delivering more accurate predictions compared to PCA-based benchmarks. In addition, the structural-break analysis identifies a break pattern in FRED-MD that is consistent with Chen et al. (2014).

Overall, our findings show that higher-order multi-cumulants provide valuable information beyond covariance in weak, non-Gaussian factor models, and that HFA offers a practical and theoretically grounded alternative to PCA in such empirically relevant settings. The remainder of this paper is organized as follows. Section 2 provides the non-Gaussian factor model assumed in HFA and the notation. Section 3 presents the tests to determine the number of factors. Section 4 introduces the HFA estimates and their properties. Section 5 presents the Adaptive HFA algorithm, which makes data-driven decisions based on tests for weak factors, non-Gaussianity and factor strength to select between PCA and HFA and determine the cumulant order. Subsequently, Section 6 reports our simulation experiments and findings, and Section 7 presents an application to market premium prediction. Finally, Section 8 provides the concluding remarks. Furthermore, additional details about this study are in our online Supplementary Appendix, and the R code is publicly available in the `hofa` package.

2 Non-Gaussian factor model

Throughout the paper, we use $\sigma_r(A)$ to denote the r-th largest singular value of the real-valued matrix A and $\psi_r(B)$ to represent the r-th largest eigenvalue of the positive semi-definite matrix B. The Frobenius norm of a matrix A is denoted by $\|A\| = \sqrt{\mathrm{tr}(A'A)}$, where $\mathrm{tr}(\cdot)$ is the trace of a matrix. We use the notation $a \asymp b$ to denote $a = O(b)$ and $b = O(a)$. We use $a \ll b$ to denote $a = o(b)$ and $a \gg b$ to denote $b = o(a)$. $\lesssim$ (or $\gtrsim$) denotes $\leqslant$ (or $\geqslant$) up to a positive constant. $A^{\otimes k} = A \underbrace{\otimes \cdots \otimes}_{k} A$, where $\otimes$ is the Kronecker product. $[z]$ is the integer part of a real number z.

2.1 Motivation and model setup

This section introduces the main model that will be used throughout the paper. We also present several illustrative examples from economics and finance to substantiate the motivation of the paper. As in Risk et al. (2019), we suppose that the non-Gaussian observations $x_t \in \mathbb{R}^N, (t = 1, \ldots, T)$ are generated by the non-Gaussian factor model: x_t is the sum of a non-Gaussian signal c_t and a Gaussian error e_t, where the non-Gaussian signal c_t is obtained via a linear transformation of an R-dimensional factor f_t:

$$x_t = c_t + e_t = \Lambda f_t + e_t \,, \tag{1}$$

where $\Lambda \in \mathbb{R}^{N \times R}$ is the factor loading matrix, and $e_t = G_N^{1/2} u_t$ with $u_t \sim \mathcal{N}(0, \mathbf{I}_N)$, where $\mathbf{I}_N$ denotes the $N \times N$ identity matrix and G_N is an $N \times N$ positive definite matrix. The factors, factor loadings, and idiosyncratic errors are not observable. While Risk et al. (2019) assume that the elements of the factors f_t are mutually independent, we allow for dependence among them. Following Bai and Ng (2002), we treat the entries in Λ as parameters and those in f_t as random variables.

Remark 1. The exploration and exploitation of non-Gaussian features in variables have proven valuable across multiple disciplines. Notable examples include independent component analysis in signal processing (Hyvärinen, 1999), structural autoregressive models in causal inference (Hyvärinen et al., 2010), and higher-order factor analysis in psychology (Mooijaart, 1985). The proposed non-Gaussian factor model in (1) assumes that all non-normality in x_t comes from the linear exposure to the factors f_t, which implies that some factors f_t are non-Gaussian. We present two examples below to highlight that nonlinear dependence among latent factors serves as a mechanism through which linear non-Gaussian factor models may arise. Figure 6 shows that the factors extracted from the FRED-MD dataset used in our empirical application exhibit pronounced departures from normality.

Example 1 (**Polynomial structural equation model**)**.** The polynomial structural equation models with three factors in Wall and Amemiya (2000) can be written as follows:

$$x_t = \Lambda f_t + e_t \,, \qquad f_{1t} = \alpha' h(f_{2t}, f_{3t}) + \zeta_t \,,$$

where $f_t = (f_{1t}, f_{2t}, f_{3t})'$, and $h(f_{2t}, f_{3t})$ is an $h \times 1$ vector with each component being a pure or mixed power of f_{2t} and f_{3t}, α is an $h \times 1$ unknown coefficient vector, and ζ_t is the measurement error between factors. Wall and Amemiya (2000) mention a simple quadratic model ($f_{1t} = \alpha_0 + \alpha_1 f_{2t} + \alpha_2 f_{2t}^2 + \zeta_t$) and a simple cross-product model ($f_{1t} =$

$\alpha_0 + \alpha_1 f_{2t} + \alpha_2 f_{3t} + \alpha_3 f_{2t} f_{3t} + \zeta_t$) as two special cases. Note that, even if f_{2t} and f_{3t} are Gaussian, the nonlinear dependence between them can lead to a non-Gaussian factor f_{1t} and thus to non-Gaussianity in x_t. The quadratic market model proposed by Barone Adesi et al. (2004) is also a special case of the polynomial structural equation model of Wall and Amemiya (2000). In this model, x_t denotes the stock return, and the factor vector is given by $f_t = (r_{M,t}, r_{M,t}^2)'$, where $r_{M,t}$ represents the market return. The motivation for including the square of the market return is to fully account for co-skewness between the stock return and the market return.

Example 2 (**Structural changes**)**.** The non-normality of factors can arise from structural breaks within factor models (Chen et al., 2014; Ma and Tu, 2023). Consider a single factor model with a structural break:

$$x_t = \begin{cases} \lambda_1^* f_t^* + e_t \, , \; t = 1, \dots, [T/2] \, , \\ \lambda_2^* f_t^* + e_t \, , \; t = [T/2] + 1, \dots, T \, , \end{cases} \tag{2}$$

where λ_1^*, λ_2^* are the factor loadings before and after the break. Following Chen et al. (2014), model (2) can be equivalently represented as a two-factor model:

$$x_t = \lambda_1^* f_{1t} + (\lambda_2^* - \lambda_1^*) f_{2t} + e_t = \Lambda f_t + e_t \, , \tag{3}$$

where $f_{1t} = f_t^*$, $f_{2t} = 0$ for $t = 1, \dots, [T/2]$ and $f_{2t} = f_t^*$ for $t = [T/2] + 1, \dots, T$, $f_t = (f_{1t}, f_{2t})'$ and $\Lambda = (\lambda_1^*, \lambda_2^* - \lambda_1^*)$. Note that, even if f_t^* follows a normal distribution, f_t is inherently non-Gaussian due to the structural break at $[T/2]$. The above arguments likewise hold in the threshold factor model (Liu and Chen, 2020) and the regime-switching factor model (Barigozzi and Massacci, 2025).

Our contribution is to identify the factor structure using information from higher-order multi-cumulants. The multi-cumulants for the second-, third-, and fourth-order moments of $z_t = (z_{1t}, z_{2t}, \dots, z_{Qt}) \in \mathbb{R}^{Q \times 1}$ with zero mean are

$$\begin{gathered} \kappa_{z,i_1 i_2,t} = m_{z,i_1 i_2,t} \, , \qquad \kappa_{z,i_1 i_2 i_3,t} = m_{z,i_1 i_2 i_3,t} \, , \\ \kappa_{z,i_1 i_2 i_3 i_4,t} = m_{z,i_1 i_2 i_3 i_4,t} - m_{z,i_1 i_2,t} m_{z,i_3 i_4,t} - m_{z,i_1 i_3,t} m_{z,i_2 i_4,t} - m_{z,i_1 i_4,t} m_{z,i_2 i_3,t} \, , \end{gathered} \tag{4}$$

where $\kappa_{z,i_1 i_2 \dots i_k,t}$ is the k-th order multi-cumulant of $z_{i_1 t}, z_{i_2 t}, \dots, z_{i_k t}$, $i_q \in \{1, 2, \dots, Q\}$ for $1 \leqslant q \leqslant k$ and $k = 2, 3, 4$, and $m_{z,i_1 i_2 \dots i_k,t} = \mathbb{E}(z_{i_1 t} z_{i_2 t} \dots z_{i_k t})$. In this paper, z_t represents either x_t, f_t, or e_t. Following Kolda and Bader (2009), we can express the k-th order multi-cumulant tensor of z_t in matrix form:

$$\mathbf{C}_{z,t}^{(k)} = \{\kappa_{z,i_1 j,t}\} \in \mathbb{R}^{Q \times Q^{k-1}} , \tag{5}$$

where $j = 1 + \sum_{p=2}^{k}(i_p - 1)Q^{p-2}$. We define

$$\mathbf{C}_{z}^{(k)} \equiv \lim_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T} \mathbf{C}_{z,t}^{(k)} \tag{6}$$

if the constant matrix $\mathbf{C}_z^{(k)}$ exists. The matrix $\mathbf{C}_z^{(k)}$ is defined as the (population) k-th order multi-cumulant for $\{z_t\}_{t=1}^T$ as $T \to \infty$. Furthermore, for the available observations $\{z_t\}_{t=1}^T$, we define the sample k-th order multi-cumulant as

$$\widetilde{\mathbf{C}}_{z}^{(k)} \equiv \frac{1}{T}\sum_{t=1}^{T} \widetilde{\mathbf{C}}_{z,t}^{(k)} . \tag{7}$$

where $\widetilde{\mathbf{C}}_{z,t}^{(k)}$ is defined analogously, using the notation in (4) and (5) and replacing $m_{z,i_1 i_2 \ldots i_k,t}$ with $\widetilde{m}_{z,i_1 i_2 \ldots i_k,t} = z_{i_1 t} z_{i_2 t} \ldots z_{i_k t}$.

To ensure that factors can be successfully identified by HFA, we impose the following assumptions on the factors:

Assumption 1. (**The factors**)

(i) $\mathbb{E}\|f_t\|^{2K} < C_1$ for a real integer $K \geqslant 3$, where C_1 is a finite constant.

(ii) For the sequence $\{f_t\}_{t=1}^T$, its k-th order sample multi-cumulant ($2 \leqslant k \leqslant K$) defined in (7) satisfies $\widetilde{\mathbf{C}}_f^{(k)} \xrightarrow{\text{p}} \mathbf{C}_f^{(k)}$ as $T \to \infty$, where $\mathbf{C}_f^{(k)}$ is a constant matrix defined in (6). Let $\phi_r^{(k)} = \sigma_r(\mathbf{C}_f^{(k)})$, $0 < \phi_r^{(k)} < \infty$ for $2 \leqslant k \leqslant K$ and $r = 1, 2, \ldots, R$.

(iii) The variables f_t and e_t are mutually independent.

Assumption 1(i) is common in factor analysis. Assumption 1(ii) requires that the sample k-th order multi-cumulant of the sequence $\{f_t\}_{t=1}^T$ converges to a constant matrix as $T \to \infty$. We further assume that all singular values of $\mathbf{C}_f^{(k)}$ are non-zero for $2 \leqslant k \leqslant K$. This assumption is imposed for the sake of simplifying the analysis; in fact, the k-th order HFA only requires that $\mathbf{C}_f^{(k)}$ satisfies the corresponding conditions. For the two examples in Remark 1, we demonstrate that this assumption holds for some $k \geqslant 3$. Assumption 1(iii) requires exogenous idiosyncratic errors. This assumption is common in latent factor modeling (see, for example, Boudt et al. 2020; Zhang et al. 2024). The assumption is needed to derive the theoretical properties of the GER and HFA approaches.

Example 1 (Continued). For the quadratic market model, let $f_t = (r_{M,t}, r_{M,t}^2)'$ and assume simply that $r_{M,t} \sim i.i.d.\,\mathcal{N}(0,1)$. The third and fourth-order multi-cumulants have the following forms:

$$\mathbf{C}_f^{(3)} = \begin{pmatrix} 0 & 2 & 2 & 0 \\ 2 & 0 & 0 & 8 \end{pmatrix}, \quad \mathbf{C}_f^{(4)} = \begin{pmatrix} 0 & 0 & 0 & 8 & 0 & 8 & 8 & 0 \\ 0 & 8 & 8 & 0 & 8 & 0 & 0 & 48 \end{pmatrix}.$$

The singular values of $\mathbf{C}_f^{(3)}$ and $\mathbf{C}_f^{(4)}$ are all nonzero, which implies that both $\mathbf{C}_f^{(3)}$ and $\mathbf{C}_f^{(4)}$ are of full rank. If $r_{M,t}$ is non-Gaussian, then $\mathbf{C}_f^{(3)}$ or $\mathbf{C}_f^{(4)}$ is obviously of full rank.

Example 2 (Continued). For the factor model with a structural break, let $f_t^* \sim i.i.d.\,\mathcal{N}(0,1)$. The third-order multi-cumulant of $f_t = (f_{1t}, f_{2t})'$ is a matrix of zeros. Its fourth-order multi-cumulant has the following form:

$$\mathbf{C}_f^{(4)} = \begin{pmatrix} 0 & 0 & 0 & 1/2 & 0 & 1/2 & 1/2 & 3/4 \\ 0 & 1/2 & 1/2 & 3/4 & 1/2 & 3/4 & 3/4 & 3/4 \end{pmatrix}.$$

The singular values $\sigma_1(\mathbf{C}_f^{(4)}) \approx 2.01$ and $\sigma_2(\mathbf{C}_f^{(4)}) \approx 0.52$, which implies that $\mathbf{C}_f^{(4)}$ is of full rank. If f_t^* is non-Gaussian, then $\mathbf{C}_f^{(3)}$ or $\mathbf{C}_f^{(4)}$ is obviously of full rank.

For the factor loadings, we adopt the heterogeneous factor strength setting of Bai and Ng (2023). Let $0 \leqslant \alpha_1 \leqslant \alpha_2 \leqslant \cdots \leqslant \alpha_R \leqslant 1$ and define the $R \times R$ normalization matrix $B_N = \mathrm{diag}(N^{(1-\alpha_1)/2}, \ldots, N^{(1-\alpha_R)/2})$. The assumptions on factor loadings are as follows:

Assumption 2. (**The factor loadings**)

(i) $\lim_{N\to\infty} B_N^{-1/2} \Lambda' \Lambda B_N^{-1/2} = \Sigma_\Lambda$ for some constant positive definite matrix Σ_Λ.

(ii) Denote $\Lambda = (\lambda_1, \ldots, \lambda_N)'$, where each λ_i is an R-dimensional vector. We have $\|\lambda_i\| < C_2$ for all $i = 1, 2, \ldots, N$ with a finite constant C_2.

Write $\Lambda = (\lambda_{ij})_{N\times R}$. Assumption 2 implies that $\sum_{i=1}^N \lambda_{ij}^2 \asymp O(N^{1-\alpha_j})$ for $j = 1, \ldots, R$, which aligns with the heterogeneous factor strength assumption of Freyaldenhoven (2022), Uematsu and Yamagata (2022) and Bai and Ng (2023). Here, α_1 represents the strongest factor strength and α_R the weakest. Onatski (2012) discusses the special case where $\alpha_1 = \cdots = \alpha_R = 1$. Bai and Ng (2023) and Uematsu and Yamagata (2022) require $\alpha_R < 1$ and Freyaldenhoven (2022) assumes $\alpha_R < 1/2$. Next, we show that the weak factor assumption is valid for the FRED-MD variables that we study in our empirical application.

Remark 2. Uematsu and Yamagata (2022) proposed a sparsity-induced weak factor model, where the j-th column of the factor loading matrix Λ contains $N_j = N^{1-\alpha_j}$ non-zero entries. In this framework, α_j equals $1 - \log N_j / \log N$, with N_j the number of non-zero loadings for the j-th factor. Bailey et al. (2021) use the classical OLS estimates of the factor loadings and compute the significant loadings by comparing the t-test statistic with a critical value function. Figure 1(i) shows the estimated factor strength obtained using Bailey et al. (2021)'s recommended implementation for the first three PCA factors based on the FRED-MD database. The results reveal that the factor strengths range from 0.1 to 0.3, which implies that all factors are weak. We obtain similar results when using the sparse orthogonal factor regression (SOFAR) estimator of Uematsu and Yamagata (2022), see Figure 1(ii).

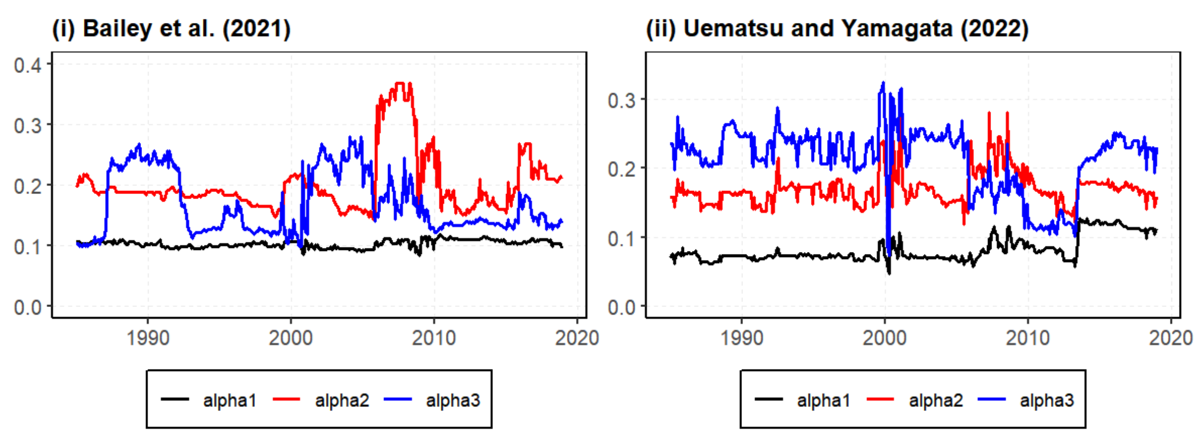

Figure 1: Estimated factor strengths for the 124 FRED-MD variables

Remark 3. In structural equation models, latent factors in psychometrics, economics, and finance typically correspond to distinct psychological constructs, economic sectors, industry segments, risk sources, or transmission mechanisms. Consequently, the loading matrix Λ

in Example 1 often exhibits a block-wise or sparse structure, which leads to weak factor loadings; this is well captured by the grouped factor model (Zhang et al., 2024). For the factor model with a structural break, the two-factor representation in (3) also implies sparsity when the break affects only a subset of variables. In this case, for unaffected variables the component $\lambda_1^* - \lambda_2^*$ in $\Lambda = (\lambda_1^*, \lambda_1^* - \lambda_2^*)$ equals zero, which results in a representation of the weak factor model.

The assumptions on the idiosyncratic errors are as follows:

Assumption 3. (**The idiosyncratic errors**)
Let $e_t = G_N^{1/2} u_t = (e_{1t}, \ldots, e_{Nt})$ for $t = 1, \ldots, T$, where $u_t = (u_{1t}, \ldots, u_{Nt}) \in \mathbb{R}^N$ and $G_N^{1/2}$ is the symmetric square root of a $N \times N$ matrix G_N.

(i) For each i, $\{u_{it}\}_{t=1}^T$ is a strong-mixing Gaussian sequence such that $\mathbb{E}(u_{it}) = 0$ and $\mathbb{E}(u_{it}^2) = 1$. The mixing coefficients $\bar{\alpha}_i(\cdot)$ for $\{u_{it}\}_{t=1}^T$ satisfy $\max_i \bar{\alpha}_i(n) \leqslant C_{\bar{\alpha}} \tau^n$ for some $C_{\bar{\alpha}} > 0$ and $\tau \in (0, 1)$. $\{u_{it}\}_{t=1}^T$ are independent across i.

(ii) $\psi_1(G_N) < C_3$ uniformly in N with a finite constant C_3.

Assumption 3 models the idiosyncratic errors $\{e_t\}_{t=1}^T$ as a weighted combination of the strong mixing sequence $\{u_t\}_{t=1}^T$. The specification allows accommodation for serial and cross-sectional correlation in the idiosyncratic errors. The strong mixing setting is common to characterize serial dependence of time series, see e.g. Su et al. (2016) and Chang et al. (2018). The matrix G_N is used to characterize the cross-sectional dependence of the error terms.

Remark 4. The assumption of Gaussian error terms is pervasive in linear factor models (Onatski, 2012; Risk et al., 2019). One driver of normality is that, as in Onatski (2012), Ahn and Horenstein (2013), and Zhang et al. (2024), we specify the idiosyncratic error e_t as a linear transformation of independent latent variables u_t (i.e., $e_t = G_N^{1/2} u_t$). Seen from this angle, the aggregation dampens some of the potential non-normality in u_t versus e_t, as discussed, for example, in Theorem 3.1 of Granger (1976). In this paper, we use normality of the error term as a sufficient condition to derive the properties of the proposed HFA methodology. In Section 5.1, we demonstrate that, when some error terms deviate from normality, the GER and HFA estimators can still reliably recover the underlying factors under certain conditions on factor strengths.

Remark 5. Assumption 3(i) follows Su et al. (2016)'s Assumption A1(i). This strong mixing assumption is needed for all asymptotic results presented in this paper. Write $E = (e_{it})_{T\times N}$. Onatski (2010) and Ahn and Horenstein (2013) specify $E = R_T^{1/2} U^* G_N^{1/2}$, where $U^* = (u^*_{it})_{T\times N}$ are *i.i.d.* standard normal random variables, $R_T^{1/2}$ and $G_N^{1/2}$ are the symmetric square root of $T \times T$ and $N \times N$ positive semidefinite matrices R_T and G_N, respectively. We thus have a different approach to describing the serial dependence, which has the following properties: (i) we allow heterogeneity in the serial dependence of $\{e_{it}\}_{t=1}^T$ and characterize it using the mixing coefficient $\bar{\alpha}_i(\cdot)$; (ii) the serial dependence is characterized by $\max_i \bar{\alpha}_i(n) \leqslant C_{\bar{\alpha}} \tau^n$, which is a sufficient condition for the assumption $\psi_1(R_T) < \infty$ in Onatski (2010) and Ahn and Horenstein (2013).

2.2 Identification of the factor loading matrix

In this section, we first derive the population covariance and higher-order multi-cumulants of x_t as implied by Assumptions 1 – 3. Then we show that the columns of the factor loading matrix Λ in (1) are the eigenvectors of $\mathbf{C}_x^{(k)} \mathbf{C}_x^{(k)\prime}$ $(3 \leqslant k \leqslant K)$ up to an invertible matrix.

First, based on the factor model (1) and Assumptions 1 and 2, the k-th order multi-cumulant of $x_{i_1 t}, x_{i_2 t}, \ldots, x_{i_k t}$ $(3 \leqslant k \leqslant K)$ with $i_m \in \{1, 2, \ldots, N\}$ and $m \in \{1, \ldots, k\}$, can be expressed as

$$\kappa_{x,i_1 \ldots i_k,t} = \lambda'_{i_1} \mathbf{C}_{f,t}^{(k)} (\lambda_{i_2} \otimes \ldots \otimes \lambda_{i_k}) + \kappa_{e,i_1 \ldots i_k,t}\,, \tag{8}$$

where $\mathbf{C}_{f,t}^{(k)}$ is the k-th order multi-cumulant matrix of f_t defined in (5), and $\kappa_{e,i_1 \ldots i_k,t}$ is the k-th order multi-cumulant of $e_{i_1 t}, e_{i_2 t}, \ldots, e_{i_k t}$. By Assumption 3, the idiosyncratic errors e_t are Gaussian-distributed. It follows that $\kappa_{e,i_1 \ldots i_k,t} = 0$ for $i_m \in \{1, 2, \ldots, N\}$ and $m \in \{1, \ldots, k\}$. Therefore, we can rewrite (8) in matrix form as $\mathbf{C}_{x,t}^{(k)} = \Lambda \mathbf{C}_{f,t}^{(k)} (\Lambda'^{\otimes(k-1)})$. For a given N and $T \to \infty$, under Assumptions 1 - 3, the population k-th order multi-cumulant $\mathbf{C}_x^{(k)}$

$(3 \leqslant k \leqslant K)$ can be expressed as

$$\mathbf{C}_x^{(k)} = \lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \mathbf{C}_{x,t}^{(k)} = \Lambda \mathbf{C}_f^{(k)} (\Lambda'^{\otimes(k-1)}) \,, \tag{9}$$

where $\mathbf{C}_f^{(k)} = \lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \mathbf{C}_{f,t}^{(k)}$ defined in (6) is the population k-th order multi-cumulant matrix of f_t. The factor f_t and the factor loading matrix Λ are not uniquely determined by (1): for a given k, we obtain the same higher-order multi-cumulant when we replace (Λ, f_t) by $(\Lambda^{(k)}, f_t^{(k)})$ such that $f_t^{(k)} = (H^{(k)})^{-1} f_t$ and $\Lambda^{(k)} = \Lambda H^{(k)}$ for an invertible matrix $H^{(k)}$. Hence, for a given k, we impose the normalization restrictions

$$\frac{1}{N} \Lambda^{(k)\prime} \Lambda^{(k)} = \mathbf{I}_R \,, \quad \mathbf{C}_{f,k}^{(k)} \mathbf{C}_{f,k}^{(k)\prime} \quad \text{is diagonal} \,, \tag{10}$$

where $\mathbf{C}_{f,k}^{(k)}$ is defined in the same manner as $\mathbf{C}_f^{(k)}$ in (6) but with replacing f_t by $f_t^{(k)}$. Under Assumptions 1 - 3, together with (9) and (10), for $3 \leqslant k \leqslant K$, it follows that

$$\begin{aligned} \mathbf{C}_x^{(k)} \mathbf{C}_x^{(k)\prime} &= \Lambda^{(k)} \mathbf{C}_{f,k}^{(k)} (\Lambda^{(k)\prime} \Lambda^{(k)})^{\otimes(k-1)} \mathbf{C}_{f,k}^{(k)\prime} \Lambda^{(k)\prime} \\ &= \Lambda^{(k)} (N^{k-1} \mathbf{C}_{f,k}^{(k)} \mathbf{C}_{f,k}^{(k)\prime}) \Lambda^{(k)\prime} \,. \end{aligned} \tag{11}$$

Therefore, (11) implies that $\Lambda^{(k)}$ is $\sqrt{N}$ times the eigenvectors corresponding to the R largest eigenvalues of $\mathbf{C}_x^{(k)} \mathbf{C}_x^{(k)\prime}$. We will use this result to obtain the HFA estimates in Section 4.1. Note that the result in (11) does not hold for $k = 2$ since $\mathbf{C}_x^{(2)} = \Lambda \mathbf{C}_f^{(2)} \Lambda' + \mathbf{C}_e^{(2)}$, where $\mathbf{C}_e^{(2)}$ is the covariance matrix of e_t.

3 Estimation of the number of factors

3.1 Generalized eigenvalue ratio estimator

We propose to estimate the number of factors R based on the sample higher-order multi-cumulant of the data $\{x_t\}_{t=1}^T$. For $r = 1, 2, \ldots, N$, we define $\widetilde{\mu}_{NT,r}^{(k)}$ as the square of the r-th largest singular value of $\widetilde{\mathbf{C}}_x^{(k)}$ (or equivalently as the r-th largest eigenvalue of $\widetilde{\mathbf{C}}_x^{(k)} \widetilde{\mathbf{C}}_x^{(k)\prime}$):

$$\widetilde{\mu}_{NT,r}^{(k)} \equiv \sigma_r^2(\widetilde{\mathbf{C}}_x^{(k)}) = \psi_r(\widetilde{\mathbf{C}}_x^{(k)} \widetilde{\mathbf{C}}_x^{(k)\prime}) \,,$$

where the sample k-th order multi-cumulant $\widetilde{\mathbf{C}}_x^{(k)}$ is defined in (7). We propose to determine the number of factors by maximizing the ratio of the two adjacent eigenvalues of $\widetilde{\mathbf{C}}_x^{(k)} \widetilde{\mathbf{C}}_x^{(k)\prime}$.

The corresponding criterion function is defined as follows:

$$\mathrm{GER}^{(k)}(r) \equiv \frac{\widetilde{\mu}^{(k)}_{NT,r}}{\widetilde{\mu}^{(k)}_{NT,r+1}}\,, \quad r = 1, 2, \ldots, R_{\max}\,,$$

where $R_{\max}$ denotes the maximum possible number of factors and $3 \leqslant k \leqslant K$, with K being the maximum order considered. We call it the GER estimator because it is an extension of Ahn and Horenstein (2013)'s eigenvalue ratio (ER) estimator. Our proposed estimator for R is the maximizer of $\mathrm{GER}^{(k)}(r)$:

$$\hat{R}^{(k)}_{\mathrm{GER}} = \max_{1 \leqslant r \leqslant R_{\max}} \mathrm{GER}^{(k)}(r)\,.$$

Our main theoretical results for selecting the number of factors are as follows.

Theorem 1. Suppose that Assumptions 1 – 3 hold with $R \geqslant 1$.

(i) If $\alpha_1 = \alpha_2 = \cdots = \alpha_R = \alpha$, and $N^{\alpha}/T \to 0$ as $(N,T) \to \infty$, and $\mathrm{tr}(G_N) = o\big(N^{\frac{k}{k-1}(1-\alpha)} T^{\frac{1}{k-1}} (\log N)^{-\frac{1}{k-1}}\big)$, then,

$$\lim_{(N,T)\to\infty} \mathbb{P}(\hat{R}^{(k)}_{\mathrm{GER}} = R) = 1$$

for any $3 \leqslant k \leqslant K$, where $R_{\max} \in (R, N-R-1]$.

(ii) If the factors f_t are mutually independent for all t, and $N^{(\alpha_R-\alpha_1)(2k-1)+\alpha_R}/T \to 0$ as $(N,T) \to \infty$, and $\mathrm{tr}(G_N) = o\big(N^{\frac{2k}{k-1}(\alpha_1-\alpha_R)} N^{\frac{k}{k-1}(1-\alpha_1)} T^{\frac{1}{k-1}} (\log N)^{-\frac{1}{k-1}}\big)$, then,

$$\lim_{(N,T)\to\infty} \mathbb{P}(\hat{R}^{(k)}_{\mathrm{GER}} = R) = 1$$

for any $3 \leqslant k \leqslant K$, where $R_{\max} \in (R, N-R-1]$.

Theorem 1 establishes the consistency of the GER estimator under two scenarios. Scenario (i) corresponds to the homogeneous factor strength case, as assumed by Bai and Ng (2023) and Zhang et al. (2024). Scenario (ii) allows for the heterogeneous factor strength case, but requires the assumption of factor independence. This additional assumption is consistent with Bai and Ng (2023) who require factor orthogonality for estimation of the factors based on the covariance matrix when factor strengths are heterogeneous. As our approach uses the higher-order cumulants, we require the stronger assumption of factor independence.

Remark 6. Consider the case where G_N has a polynomial rate decaying eigenvalue spectrum given by $\psi_j(G_N) \leqslant C_0 j^{-\rho}$ for some $\rho \geqslant 0$. We refer to ρ as "the spectrum decay rate" as in Li et al. (2021). Then $\mathrm{tr}(G_N) = O(N^{1-\rho})$ for $0 \leqslant \rho < 1$, $\mathrm{tr}(G_N) = O(\log N)$ for $\rho = 1$ and $\mathrm{tr}(G_N) = O(1)$ for some $\rho > 1$. In the Supplementary Appendix, we show that this assumption is validated for the dataset used in the empirical analysis.

Remark 7. (i) The parameter K denotes the maximum order considered. As a rule of thumb, we recommend setting $K = 4$; throughout this paper we focus on $k \in \{3, 4\}$. (ii) We recommend setting $R_{\max}$ as in Ahn and Horenstein (2013). See the Supplementary Appendix for details. (iii) Ahn and Horenstein (2013) proposes a growth ratio estimator, which generalizes the ER estimator for the number of factors. In the Supplementary Appendix, we develop a higher-order-cumulant analogue, termed the Generalized Growth Ratio (GGR) estimator.

3.2 GER versus ER

In this section, we use a stylized setup to show that the GER estimator has better finite sample properties than Ahn and Horenstein (2013)'s ER estimator for detecting weak factors. We consider a two-factor model $x_{it} = \lambda_{i1} f_{1t} + \lambda_{i2} f_{2t} + e_{it}$ with different factor strengths α_1 and α_2. Specifically, the factor loadings satisfy $\sum_{i=1}^{N} \lambda_{i1}^2 = O\big(N^{1-\alpha_1}\big)$ and $\sum_{i=1}^{N} \lambda_{i2}^2 = O\big(N^{1-\alpha_2}\big)$, with $\alpha_1 \leqslant \alpha_2$. Since a smaller α indicates a stronger factor, it follows that f_{1t} has greater strength than f_{2t}. Suppose that Assumptions 1 – 3 hold. Additionally, $\psi_j(G_N) = C_0 j^{-\rho}$ with $\rho \geqslant 0$ and some positive constant C_0 for $j = 1, 2, \ldots, N$.

For a given N, we first study the power of the ER estimator based on the eigenvalue decomposition of the sample covariance matrix of x_t, labeled as $\widetilde{\Sigma}_{x,N}$. Following the arguments in Ahn and Horenstein (2013), we can show that

$$\psi_1(\widetilde{\Sigma}_{x,N}) / \psi_2(\widetilde{\Sigma}_{x,N}) \asymp O_p(N^{\alpha_2 - \alpha_1}) \,,$$
$$\psi_2(\widetilde{\Sigma}_{x,N}) / \psi_3(\widetilde{\Sigma}_{x,N}) \asymp O_p(N^{1-\alpha_2}) \,.$$

When $\alpha_1 = \alpha_2 = 0$, we can detect two factors because $\psi_2(\widetilde{\Sigma}_{x,N}) / \psi_3(\widetilde{\Sigma}_{x,N}) \to \infty$ as $N \to \infty$. However, when $\alpha_2 > 1/2 + \alpha_1/2$ and $\alpha_1 < 1$, it holds that $\psi_1(\widetilde{\Sigma}_{x,N}) / \psi_2(\widetilde{\Sigma}_{x,N}) \gg \psi_2(\widetilde{\Sigma}_{x,N}) / \psi_3(\widetilde{\Sigma}_{x,N})$, which implies that only f_{1t} can be detected. Therefore, the ER estimator has a low efficiency in the weak factor model because it can detect only f_{1t}.

Consider now the GER estimator based on the eigenvalue decomposition of $\widetilde{\mathbf{C}}_{x,N}^{(3)}\widetilde{\mathbf{C}}_{x,N}^{(3)\prime}$, where $\widetilde{\mathbf{C}}_{x,N}^{(3)}$ is the sample third-order multi-cumulant of x_t for a given N. We can show that

$$\begin{aligned}
&\psi_1(\widetilde{\mathbf{C}}_{x,N}^{(3)}\widetilde{\mathbf{C}}_{x,N}^{(3)\prime})/\psi_2(\widetilde{\mathbf{C}}_{x,N}^{(3)}\widetilde{\mathbf{C}}_{x,N}^{(3)\prime}) \asymp O_p(N^{\alpha_2-\alpha_1}),\\
&\psi_2(\widetilde{\mathbf{C}}_{x,N}^{(3)}\widetilde{\mathbf{C}}_{x,N}^{(3)\prime})/\psi_3(\widetilde{\mathbf{C}}_{x,N}^{(3)}\widetilde{\mathbf{C}}_{x,N}^{(3)\prime}) \asymp O_p(TN^{1+2\rho-3\alpha_2}) + O_p(TN^{-\alpha_2}).
\end{aligned}$$

Here we ignore the logarithmic term $\log N$, which has negligible impact on the eigenvalue ratios of $\widetilde{\mathbf{C}}_{x,N}^{(3)}\widetilde{\mathbf{C}}_{x,N}^{(3)\prime}$. As long as the sample size (N,T) satisfies $\min(TN^{1+2\rho-4\alpha_2+\alpha_1}, TN^{\alpha_1-2\alpha_2}) \to \infty$, it holds that $\psi_1(\widetilde{\mathbf{C}}_{x,N}^{(3)}\widetilde{\mathbf{C}}_{x,N}^{(3)\prime})/\psi_2(\widetilde{\mathbf{C}}_{x,N}^{(3)}\widetilde{\mathbf{C}}_{x,N}^{(3)\prime}) \ll \psi_2(\widetilde{\mathbf{C}}_{x,N}^{(3)}\widetilde{\mathbf{C}}_{x,N}^{(3)\prime})/\psi_3(\widetilde{\mathbf{C}}_{x,N}^{(3)}\widetilde{\mathbf{C}}_{x,N}^{(3)\prime})$, which implies that the GER estimator is capable of detecting weak factors for which the ER estimator performs inefficiently. As shown in Section 8 of the Supplementary Appendix, when the eigenvalues of G_N increase, the maximum eigenvalue ratio of the ER estimator shifts from the second ratio ($\hat{R}_{ER} = 2$) to the first ratio ($\hat{R}_{ER} = 1$). This indicates that only the first factor f_{1t} is detected. Conversely, the maximum eigenvalue ratio of the GER estimator remains the second ratio ($\hat{R}_{GER}^{(3)} = 2$).

4 Estimation of the factors and loadings

4.1 HFA estimators and their asymptotic properties

In this section, we present the estimators of the factors and factor loadings and show that they possess desirable asymptotic properties. We assume that the number of factors R is known (or estimated consistently). Following (11), for a given order k such that $3 \leqslant k \leqslant K$, we can estimate the factor loading matrix Λ up to a rotation matrix by using the following optimization problem:

$$\hat{\Lambda}^{(k)} = \arg\max_{\Lambda^{(k)}} \operatorname{tr}\big\{\Lambda^{(k)\prime}(\widetilde{\mathbf{C}}_x^{(k)}\widetilde{\mathbf{C}}_x^{(k)\prime})\Lambda^{(k)}\big\} \tag{12}$$

subject to the constraint $\frac{1}{N}\Lambda^{(k)\prime}\Lambda^{(k)} = \mathbf{I}_R$. Notably, (12) is the same optimization problem as finding the eigenvectors of the matrix $\widetilde{\mathbf{C}}_x^{(k)}\widetilde{\mathbf{C}}_x^{(k)\prime}$. Given $\hat{\Lambda}^{(k)}$, the factors can be obtained by least squares regression leading to $\hat{F}^{(k)} = X\hat{\Lambda}^{(k)}(\hat{\Lambda}^{(k)\prime}\hat{\Lambda}^{(k)})^{-1} = X\hat{\Lambda}^{(k)}/N$, where $X = (x_1', \ldots, x_T')' \in \mathbb{R}^{T\times N}$ is the observed panel data. Write $F = (f_1', \ldots, f_T')' \in \mathbb{R}^{T\times R}$. Then

$(\hat{F}^{(k)}, \hat{\Lambda}^{(k)})$ is the k-th order HFA estimate of (F, Λ).

Remark 8. There is a scaling mismatch between the HFA estimator $\hat{\Lambda}^{(k)}$ and the population loading matrix Λ, induced by the normalization in (10). For instance, when $\alpha_1 = \cdots = \alpha_R = \alpha$, the underlying factor loading matrix Λ satisfies $\|\Lambda\| = O(\sqrt{N^{1-\alpha}})$ by Assumption 2(ii). However, the estimated factor loading matrix satisfies $\|\hat{\Lambda}^{(k)}\| = O(\sqrt{N})$ by the constraint $\frac{1}{N}\hat{\Lambda}^{(k)\prime}\hat{\Lambda}^{(k)} = \mathbf{I}_R$. Analogously, the estimated factor $\hat{F}^{(k)}$ satisfies $\|\hat{F}^{(k)}\| = O_p(\sqrt{TN^{-\alpha}})$ and the underlying factor F satisfies $\|F\| = O_p(\sqrt{T})$. Therefore, before comparing the estimation errors, we need to make scale adjustments to the estimators. We adopt the normalization in (10) because it yields loading estimates as eigenvectors of the multi-cumulant matrix. As a reviewer noted, other normalizations are possible, and selecting the normalization in a data-driven way is an interesting direction for future research.

To investigate the consistency of the HFA estimators, as outlined in Remark 8, we first define the rescaled estimator of factors and loadings as $\bar{F}^{(k)} = \sqrt{N}\hat{F}^{(k)}B_N^{-1}$ and $\bar{\Lambda}^{(k)} = \frac{1}{\sqrt{N}}\hat{\Lambda}^{(k)}B_N$. The following theorem provides the convergence rate of $(\bar{F}^{(k)}, \bar{\Lambda}^{(k)})$.

Theorem 2. Suppose that Assumptions 1 – 3 hold.

(i) If $\alpha_1 = \alpha_2 = \cdots = \alpha_R = \alpha$, and $N^\alpha/T \to 0$ as $(N,T) \to \infty$, and $\mathrm{tr}(G_N) = o\big(T^{\frac{2}{k-1}}N^{1-\frac{k}{k-1}\alpha}(\log N)^{-\frac{1}{k-1}}\big)$, then for any $3 \leqslant k \leqslant K$, there exists an $R \times R$ invertible matrix $H^{(k)}$ for which

$$\frac{1}{\sqrt{N^{1-\alpha}}}\|\bar{\Lambda}^{(k)} - \Lambda H^{(k)}\| = O_p(\sqrt{\frac{N^\alpha}{T}}) + O_p(\sqrt{\frac{\mathrm{tr}(G_N)^{k-1}\log N}{N^{(1-\alpha)k}T}}),$$
$$\frac{1}{\sqrt{T}}\|\bar{F}^{(k)} - F(H^{(k)\prime})^{-1}\| = O_p(\frac{1}{\sqrt{N^{1-\alpha}}}) + O_p(\frac{N^\alpha}{T}).$$

(ii) If the factors f_t are mutually independent for all t, and $N^{\alpha_R+(\alpha_R-\alpha_1)(k-1)}/T \to 0$ as $(N,T) \to \infty$, and $\mathrm{tr}(G_N) = o\big(T^{\frac{2}{k-1}}N^{1-\frac{k}{k-1}\alpha_R}(\log N)^{-\frac{1}{k-1}}\big)$, then for any $3 \leqslant k \leqslant K$, there exists an $R \times R$ invertible matrix $\check{H}^{(k)}$ for which

$$\frac{1}{\sqrt{N^{1-\alpha_R}}}\|\bar{\Lambda}^{(k)} - \Lambda\check{H}^{(k)}\| = O_p(\sqrt{\frac{N^{\alpha_R+(\alpha_R-\alpha_1)(k-1)}}{T}}) + O_p(\sqrt{\frac{\mathrm{tr}(G_N)^{k-1}\log N}{N^{(1-\alpha_R)k}T}}),$$
$$\frac{1}{\sqrt{T}}\|\bar{F}^{(k)} - F(\check{H}^{(k)\prime})^{-1}\| = O_p(\frac{1}{\sqrt{N^{1-\alpha_R}}}) + O_p(\frac{N^{\alpha_R+(\alpha_R-\alpha_1)(k-1)/2}}{T}).$$

Analogous to Theorem 1, Theorem 2 presents results for two scenarios. In the homogeneous factor strength case, Theorem 2(i) shows that the convergence rates of $\bar{F}^{(k)}$ and

$\bar{\Lambda}^{(k)}$ depend on the factor strength α, the order k, and $\text{tr}(G_N)$. When $\alpha = 0$, namely a strong factor model as in Bai (2003), the convergence rates of $\bar{F}^{(k)}$ and $\bar{\Lambda}^{(k)}$ are dominated by $O_p(T^{-1/2})$ and $O_p(N^{-1/2})$. In the heterogeneous factor strength case, Theorem 2(ii) shows that the HFA estimators remain consistent. The factor strengths that matter for the convergence rate are those of the strongest α_1 and the weakest α_R, which is consistent with the result of Proposition 6 in Bai and Ng (2023).

Write $\bar{F}^{(k)} = (\bar{f}_1^{(k)'}, \ldots, \bar{f}_T^{(k)'})'$ and $\bar{\Lambda}^{(k)} = (\bar{\lambda}_1^{(k)'}, \ldots, \bar{\lambda}_N^{(k)'})'$. To derive the limit distributions of $\bar{f}_t^{(k)}$ and $\bar{\lambda}_i^{(k)}$ for $t = 1, \ldots, T$ and $i = 1, \ldots, N$, we require the following additional assumptions:

Assumption 4. (**Central limit theorem**)

(i) Fix any $k \in \{2, \ldots, K\}$ that satisfies Assumption 1 (ii). For each i, as $T \to \infty$, $\frac{1}{\sqrt{T}} \sum_{t=1}^T \zeta_t^{(k)} e_{it} \xrightarrow{d} \mathcal{N}(0, \Theta_i^{(k)})$, where $\Theta_i^{(k)} = \lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^T \sum_{s=1}^T \mathbb{E}(\zeta_t^{(k)} \zeta_s^{(k)\prime} e_{it} e_{is})$ and $\zeta_t^{(k)} \in \mathbb{R}^{R^{k-1}}$ is the t-th row of $\bar{\mathcal{H}}_f^{(k)} \in \mathbb{R}^{T \times R^{k-1}}$. The matrix $\bar{\mathcal{H}}_f^{(k)}$ is defined by $T^{-1} F' \bar{\mathcal{H}}_f^{(k)} = \widetilde{\mathbf{C}}_f^{(k)}$.

(ii) For each t, as $N \to \infty$, $B_N^{-1} \sum_{i=1}^N \lambda_i e_{it} \xrightarrow{d} \mathcal{N}(0, \Phi_t)$, where $\Phi_t = \lim_{N\to\infty} \sum_{i=1}^N \sum_{j=1}^N \mathbb{E}(B_N^{-1} \lambda_i \lambda_j' B_N^{-1} e_{it} e_{jt})$.

Assumption 4(i) requires the asymptotic normality of the cross product between $\zeta_t^{(k)}$ and e_{it}, which depends on the cumulant order k. When $k = 2$, this assumption is equivalent to Bai (2003)'s Assumption F. The explicit form of $\bar{\mathcal{H}}_f^{(k)}$ follows directly from its definition; for example, $\bar{\mathcal{H}}_f^{(2)} = F \in \mathbb{R}^{T \times R}$ and $\bar{\mathcal{H}}_f^{(3)} = (F' \odot F')' \in \mathbb{R}^{T \times R^2}$, where $\odot$ denotes the Khatri-Rao product. Notice that $\Theta_i^{(k)}$ is an $R^{k-1} \times R^{k-1}$ matrix, and Φ_t is an $R \times R$ matrix. Assumption 4(ii) requires the asymptotic normality of the cross product between λ_i and e_{it}, and the CLT for the factor loadings when $\alpha_1 = \cdots = \alpha_R = 0$ has the same form as in Bai (2003)'s Assumption F. If $\alpha_R > 0$, we allow for a slower convergence rate than $\sqrt{N}$. Subsequently, the following result holds:

Theorem 3. Suppose that Assumptions 1 – 4 hold.

(i) If $\alpha_1 = \alpha_2 = \cdots = \alpha_R = \alpha$, then for any $3 \leqslant k \leqslant K$, we have

$$\sqrt{T}\big(\bar{\lambda}_i^{(k)} - H^{(k)'} \lambda_i\big) \xrightarrow{d} \mathcal{N}\Big(0, (D^{(k)})^{-1} Q^{(k)'} \mathbf{C}_f^{(k)} (\Sigma_\Lambda)^{\otimes k-1} \Theta_i^{(k)} (\Sigma_\Lambda)^{\otimes k-1} \mathbf{C}_f^{(k)'} Q^{(k)} (D^{(k)})^{-1}\Big),$$

provided $N^{\alpha}/T = o(1)$ and $\mathrm{tr}(G_N) = o\big(N^{\frac{k}{k-1}-\alpha}(\log N)^{-\frac{1}{k-1}}\big)$, where $D^{(k)} = p\lim_{(N,T)\to\infty} \widetilde{D}_{N,T}^{(k)}$ and $\widetilde{D}_{N,T}^{(k)}$ is an $R\times R$ diagonal matrix of the first R largest eigenvalues of $\frac{1}{N^{(1-\alpha)k}}\widetilde{\mathbf{C}}_x^{(k)}\widetilde{\mathbf{C}}_x^{(k)\prime}$ in decreasing order, and $Q^{(k)} = p\lim_{(N,T)\to\infty} \Lambda'\bar{\Lambda}^{(k)}/N^{1-\alpha}$. Further,

$$\sqrt{N^{1-\alpha}}\big(\bar{f}_t^{(k)} - (H^{(k)})^{-1} f_t\big) \xrightarrow{d} \mathcal{N}\Big(0, (Q^{(k)\prime})^{-1}\Phi_t(Q^{(k)})^{-1}\Big),$$

provided $N^{\alpha}/T = o(1)$, $\alpha < 1$ and $\mathrm{tr}(G_N) = o(T^{\frac{1}{k-1}}N^{(1-\alpha)\frac{k}{k-1}}(\log N)^{-\frac{1}{k-1}})$.

(ii) If the factors f_t are mutually independent for all t, then for any $3 \leqslant k \leqslant K$, we have

$$\sqrt{T}\big(\bar{\lambda}_i^{(k)} - \check{H}^{(k)\prime}\lambda_i\big) \xrightarrow{d} \mathcal{N}\Big(0, (\bar{D}^{(k)})^{-1}\bar{Q}^{(k)\prime}\mathbf{C}_f^{(k)}(\Sigma_\Lambda)^{\otimes k-1}\Theta_i^{(k)}(\Sigma_\Lambda)^{\otimes k-1}\mathbf{C}_f^{(k)\prime}\bar{Q}^{(k)}(\bar{D}^{(k)})^{-1}\Big),$$

provided $N^{\alpha_R+2(\alpha_R-\alpha_1)(k-1)}/T = o(1)$ and $\mathrm{tr}(G_N) = o(N^{\frac{k}{k-1}-\alpha_1}(\log N)^{-\frac{1}{k-1}})$, where $\bar{D}^{(k)} = p\lim_{(N,T)\to\infty} B_N^{-2k}\check{D}_{N,T}^{(k)}$ and $\check{D}_{N,T}^{(k)}$ is an $R\times R$ diagonal matrix of the first R largest eigenvalues of $\widetilde{\mathbf{C}}_x^{(k)}\widetilde{\mathbf{C}}_x^{(k)\prime}$ in decreasing order, and $\bar{Q}^{(k)} = p\lim_{(N,T)\to\infty} B_N^{k-1}\Lambda'\bar{\Lambda}^{(k)}B_N^{-k-1}$. Further,

$$B_N\big(\bar{f}_t^{(k)} - (\check{H}^{(k)})^{-1} f_t\big) \xrightarrow{d} \mathcal{N}\Big(0, (\bar{Q}^{(k)\prime})^{-1}\Phi_t(\bar{Q}^{(k)})^{-1}\Big).$$

provided $N^{\alpha_R+(\alpha_R-\alpha_1)(k-1)}/T = o(1)$, $\alpha_R < 1$ and $\mathrm{tr}(G_N) = o\big(T^{\frac{1}{k-1}}N^{(1-\alpha_R)\frac{k}{k-1}}(\log N)^{-\frac{1}{k-1}}\big)$.

The central limit theorems in Theorem 3 have two eye-catching results. First, the estimated factor loading $\bar{\lambda}_i^{(k)}$ is asymptotically normal when $\mathrm{tr}(G_N)$, α_1 and α_R satisfy specific conditions; for example, if $\mathrm{tr}(G_N) = O(N)$, then $\bar{\lambda}_i^{(k)}$ is asymptotically normal for $\alpha_R < \frac{1}{k-1}$ if we ignore the logarithmic term. Second, the asymptotic normality of the estimated factor $\bar{f}_t^{(k)}$ holds for all $\alpha_j \in [0,1)$ under certain conditions on (N,T) and $\mathrm{tr}(G_N)$. Specifically, in the case of homogeneous factor strength, we require $N^{\alpha}/T = o(1)$ and $\mathrm{tr}(G_N) = o(T^{\frac{1}{k-1}}N^{(1-\alpha)\frac{k}{k-1}}(\log N)^{-\frac{1}{k-1}})$.

4.2 Efficiency gain of HFA estimators

We compare the efficiency of HFA and PCA in terms of convergence rates and asymptotic variances. For notational simplicity, we focus on homogeneous factor strength ($\alpha_1 = \cdots = \alpha_R = \alpha$); the heterogeneous case follows analogously. Let $(\bar{F}^{(PCA)}, \bar{\Lambda}^{(PCA)})$ denote the PCA estimators of (F,Λ), normalized by $\bar{F}^{(PCA)\prime}\bar{F}^{(PCA)}/T = \mathbf{I}_R$ and making $\bar{\Lambda}^{(PCA)\prime}\bar{\Lambda}^{(PCA)}$ diagonal. By Propositions 1–2 of Bai and Ng (2023), there exists an invertible $R\times R$ matrix

H such that

$$\frac{1}{\sqrt{N^{1-\alpha}}}\|\bar{\Lambda}^{(PCA)} - \Lambda H\| = O_p(\sqrt{\frac{N^\alpha}{T}}) + O_p(\frac{1}{N^{1-\alpha}}) ,$$
$$\frac{1}{\sqrt{T}}\|\bar{F}^{(PCA)} - F(H')^{-1}\| = O_p(\frac{1}{\sqrt{N^{1-\alpha}}}) + O_p(\frac{N^\alpha}{T}) .$$

When $\alpha = 0$, the rates reduce to the classical $O_p(T^{-1/2})$ and $O_p(N^{-1/2})$, so PCA and HFA share the same convergence rates in the strong-factor model. When $0 < \alpha < 1$, $\bar{\Lambda}^{(k)}$ converges faster than $\bar{\Lambda}^{(PCA)}$ if $\mathrm{tr}(G_N) = o(N^{(1-\alpha)\frac{k-2}{k-1}} T^{\frac{1}{k-1}} (\log N)^{-\frac{1}{k-1}})$ and $N^{2-\alpha}/T = o(1)$. Although $\bar{F}^{(k)}$ and $\bar{F}^{(PCA)}$ have the same convergence rate, $\bar{F}^{(k)}$ can be more efficient when $\bar{\Lambda}^{(k)}$ has smaller estimation error, since the factor estimates are obtained by regressing X on the corresponding loading estimates.

To compare asymptotic distributions, we eliminate the rotation effect as in Bai and Ng (2013). Write $\bar{F}^{(PCA)} = (\bar{f}_1^{(PCA)'}, \ldots, \bar{f}_T^{(PCA)'})'$ and $\bar{\Lambda}^{(PCA)} = (\bar{\lambda}_1^{(PCA)'}, \ldots, \bar{\lambda}_N^{(PCA)'})'$. In the strong-factor model ($\alpha = 0$), Bai and Ng (2013) show

$$\sqrt{T}\big(\bar{\lambda}_i^{(PCA)} - \lambda_i\big) \xrightarrow{d} \mathcal{N}\big(0, D^{-1}\Theta_i D^{-1}\big) ,$$
$$\sqrt{N}\big(\bar{f}_t^{(PCA)} - f_t\big) \xrightarrow{d} \mathcal{N}\big(0, \Phi_t\big) ,$$

where $D = p\lim_{T\to\infty} F'F/T$, $\Theta_i = p\lim_{T\to\infty} T^{-1}\sum_{t=1}^T\sum_{s=1}^T \mathbb{E}(f_t f_s' e_{it} e_{is})$, and $\Phi_t = \lim_{N\to\infty} N^{-1}\sum_{i=1}^N\sum_{j=1}^N \mathbb{E}(\lambda_i\lambda_j' e_{it} e_{jt})$. We also show that the HFA estimators satisfy

$$\sqrt{T}\big(\bar{\lambda}_i^{(k)} - \lambda_i\big) \xrightarrow{d} \mathcal{N}\big(0, (D^{(k)})^{-1}\mathbf{C}_f^{(k)}\Theta_i^{(k)}\mathbf{C}_f^{(k)'}(D^{(k)})^{-1}\big) ,$$
$$\sqrt{N}\big(\bar{f}_t^{(k)} - f_t\big) \xrightarrow{d} \mathcal{N}\big(0, \Phi_t\big) ,$$

where $[D^{(k)}]_{ii} = \sigma_i^2(\mathbf{C}_f^{(k)})$ and $\Theta_i^{(k)} = p\lim_{T\to\infty}\frac{1}{T}\sum_{t=1}^T\sum_{s=1}^T \mathbb{E}(\zeta_t^{(k)}\zeta_s^{(k)'} e_{it}e_{is})$. By Assumption 4(ii) and the definition of Φ_t, $\bar{f}_t^{(k)}$ and $\bar{f}_t^{(PCA)}$ share the same limit distribution. For factor loadings, both asymptotic variances have sandwich forms, but $\Theta_i^{(k)}$ and Θ_i differ; in particular, when $k = 2$, $\Theta_i^{(k)} = \Theta_i$, hence PCA is a special case of HFA.

In general, when e_{it} is heteroscedastic and serially correlated, $\Theta_i^{(k)}$ is hard to characterize explicitly. To highlight the variance difference, consider *i.i.d.* errors e_{it} with $\mathbb{E}(e_{it}) = 0$ and $\mathbb{E}(e_{it}^2) = \sigma_e^2$, and mutually independent factors $f_{1t}, \ldots, f_{Rt}$. Let $\sigma_j^2 \equiv \mathbb{E}(f_{jt}^2)$, $\phi_j \equiv \mathbb{E}(f_{jt}^3)$, and $\xi_j \equiv \mathbb{E}(f_{jt}^4)$. For PCA, the variance of $\sqrt{T}(\bar{\lambda}_{ij}^{(PCA)} - \lambda_{ij})$ is $[D^{-1}\Theta_i D^{-1}]_{jj} = (\sigma_j^2)^{-1}(\sigma_j^2\sigma_e^2)(\sigma_j^2)^{-1} = \sigma_e^2/\sigma_j^2$. For HFA with $k = 3$, the variance of $\sqrt{T}(\bar{\lambda}_{ij}^{(3)} - \lambda_{ij})$ is

$[(D^{(3)})^{-1}\mathbf{C}_f^{(3)}\Theta_i^{(3)}\mathbf{C}_f^{(3)\prime}(D^{(3)})^{-1}]_{jj} = \xi_j\sigma_e^2/(\phi_j)^2$. By Hölder's inequality, $\mathbb{E}(f_{jt}^2)\mathbb{E}(f_{jt}^4) \geqslant \mathbb{E}(f_{jt}^3)^2$, hence $\mathrm{Var}(\bar{\lambda}_{ij}^{(PCA)}) \leqslant \mathrm{Var}(\bar{\lambda}_{ij}^{(k)})$ in this setting, which implies the estimated factor loadings of HFA have a larger asymptotic variance compared to those of PCA.

In summary, (i) $\bar{f}_t^{(k)}$ and $\bar{f}_t^{(PCA)}$ have the same convergence rate and the same limit distribution; (ii) under suitable conditions, $\bar{\lambda}_i^{(k)}$ converges faster than $\bar{\lambda}_i^{(PCA)}$ when $\alpha > 0$; (iii) in the typical *i.i.d.* case above, $\bar{\lambda}_i^{(k)}$ has a larger asymptotic variance than $\bar{\lambda}_i^{(PCA)}$. Although $\bar{F}^{(k)}$ and $\bar{F}^{(PCA)}$ share the same rate, the simulations in Section 6.2 show that $\bar{F}^{(k)}$ achieves better finite-sample accuracy when $\alpha > 0$, consistent with the faster convergence of $\bar{\Lambda}^{(k)}$.

5 Extensions

5.1 The case of non-Gaussian idiosyncratic errors

The results of Theorems 1 – 3 are all based on the sufficient assumption that the idiosyncratic errors are normally distributed such that they have no effect on the higher-order cumulants of the data. In this section, we discuss how the assumption can be relaxed to allow for non-normality in the errors as long as their contribution to the higher-order cumulants is asymptotically negligible as $(N, T) \to \infty$. Specifically, we replace the normality requirement in Assumption 3 with the following assumption.

Assumption 5. (**Non-Gaussian idiosyncratic errors**)
Let $e_t = G_N^{1/2}u_t = (e_{1t}, \ldots, e_{Nt})$ for $t = 1, \ldots, T$, where $u_t = (u_{1t}, \ldots, u_{Nt}) \in \mathbb{R}^N$ and $G_N^{1/2}$ is the symmetric square root of an $N \times N$ matrix G_N. Let $G_N^* = \mathrm{diag}(g_1^*, \ldots, g_N^*)$ and $L = (l_1, \ldots, l_N)$, respectively, be the diagonal matrix of eigenvalues and the matrix of eigenvectors of G_N, with eigenvalues arranged in decreasing order. Let $\mathcal{G}_i = \sum_{j=1}^N \mathbf{1}\{l_{ji} \neq 0\}$ be the number of non-zero elements of $l_i = (l_{i1}, \ldots, l_{iN})'$.

(i) For each i, $\{u_{it}\}_{t=1}^T$ is a strong-mixing sequence such that $\mathbb{E}(u_{it}) = 0$, $\mathbb{E}(u_{it}^2) = 1$ and $\mathbb{E}(u_{it}^{2K}) < \infty$. The mixing coefficients $\bar{\alpha}_i(\cdot)$ for $\{u_{it}\}_{t=1}^T$ satisfy $\max_i \bar{\alpha}_i(n) \leqslant C_{\bar{\alpha}}\tau^n$ for some $C_{\bar{\alpha}} > 0$ and $\tau \in (0, 1)$. $\{u_{it}\}_{t=1}^T$ are independent across i.

(ii) $\psi_1(G_N) < C_3$ uniformly in N with a finite constant C_3.

(iii) $\kappa_{u,i}^{(k)} = \lim_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T} \kappa_{u,i,t}^{(k)}$ exists and $\sqrt{\mathcal{G}_i}|l_{ji}| = O(1)$ for those $l_{ji} \neq 0$, and

$$N^{\frac{(2\alpha_R-\alpha_1-1)k}{2}}[\mathcal{G}^{-1}\mathrm{tr}(G_N)]^{\frac{k}{2}-1}\kappa_u^{(k)} = o(1) \tag{13}$$

for any possible eigenvector matrix L of G_N, where $\mathcal{G} = \min_i \mathcal{G}_i$ and $\kappa_u^{(k)} = \max_i |\kappa_{u,i}^{(k)}|$.

Assumption 5(iii) controls the non-Gaussianity of the idiosyncratic errors. The condition (13) is automatically satisfied when $\kappa_u^{(k)} = 0$, which corresponds to the Gaussian error case in Assumption 3. When the innovation series exhibits non-normality ($\kappa_u^{(k)} > 0$) and $G_N = \mathbf{I}_N$, the condition (13) is satisfied when $\alpha_R - \alpha_1/2 < 1/k$. It follows that one should not be overly concerned about the non-normality of the errors in the case of non-Gaussian factor models with strong factors.

Proposition 1. The conclusions of Theorems 1 and 2 remain valid if Assumption 3 is replaced by Assumption 5.

We provide the complete proof in the Supplementary Appendix. Proposition 1 confirms that GER and HFA remain valid for non-Gaussian idiosyncratic errors when the contribution of $\mathbf{C}_e^{(k)}$ to higher-order multi-cumulants is asymptotically negligible. Yet empirical implementation is hindered by two key issues: Condition (13) is unverifiable (as G_N and non-Gaussianity of e_t are unobservable), and GER or HFA requires latent factors with strong non-Gaussian deviations to ensure $\mathbf{C}_f^{(k)}$ is informative. This motivates the need for a simple, data-driven guideline to judge when GER and HFA are practically reliable.

5.2 Adaptive HFA algorithm

HFA is useful when the relevant factors are weak and the non-normality of the data is mainly driven by these factors. Since such features are typically unknown a priori, we propose an Adaptive HFA (AHFA) algorithm that automates the choice between PCA and HFA. The algorithm consists of three steps.

Step 1: Test for the presence of non-Gaussian weak factors. First, extract preliminary factors using PCA and treat them as diagnostic proxies. Next, assess (i) non-normality (e.g., using the Jarque–Bera test) and (ii) factor strength (e.g., using measures in Bailey et al. (2021) or Uematsu and Yamagata (2022)). If the evidence supports the presence of weak, non-Gaussian factors, proceed to Step 2; otherwise, continue with PCA.

Step 2: Select the number of factors via the Adaptive GER estimator. When weak, non-Gaussian factors are detected, the number of factors R must be determined. For a chosen maximum order $K \geqslant 3$, the k-th order GER estimator (for $3 \leqslant k \leqslant K$) and the ER estimator are computed, and we set

$$\hat{R}_{\text{AGER}} \;=\; \max\left\{\hat{R}_{\text{GER}}^{(3)}, \ldots, \hat{R}_{\text{GER}}^{(K)},\; \hat{R}_{\text{ER}}\right\}.$$

We call $\hat{R}_{\text{AGER}}$ the Adaptive GER (AGER) estimator. Taking the maximum is a conservative choice that reduces the risk of underestimating R and omitting important factors. Section 6.3 shows that AGER performs well even when the non-normality of factors is weak.

Step 3: Choose the cumulant order and estimate factors and loadings. For $2 \leqslant k \leqslant K$, we estimate factors and factor loadings using the k-th order HFA with a pre-specified factor number $\hat{R}$ (e.g., $\hat{R} = \hat{R}_{\text{AGER}}$), where $k = 2$ corresponds to PCA. To select the cumulant order k, we choose the order that explains the largest proportion of the spectral variation in the empirical k-th order cumulant matrix. Define the factor contribution ratio (FCR) as

$$\text{FCR}(k) = \frac{\sum_{j=1}^{\hat{R}} \psi_j\big(\widetilde{\mathbf{C}}_x^{(k)} \widetilde{\mathbf{C}}_x^{(k)\prime}\big)}{\sum_{j=1}^{N} \psi_j\big(\widetilde{\mathbf{C}}_x^{(k)} \widetilde{\mathbf{C}}_x^{(k)\prime}\big)}, \qquad 2 \leqslant k \leqslant K,$$

where $\psi_j(\cdot)$ denotes the j-th largest eigenvalue. Note that FCR(2) corresponds to the proportion of variance explained by the first $\hat{R}$ PCA factors. We then set

$$k^* \;=\; \arg\max_{2 \leqslant k \leqslant K} \text{FCR}(k),$$

and define the AHFA estimator as the k^*-th order HFA estimator. Section 6.3 confirms the finite-sample performance of AHFA.

Here we briefly explain the rationale for using the FCR. For each multi-cumulant, the FCR measures the share of total variation explained by the first R common factors. When the factors are Gaussian, higher-order population cumulants vanish and their empirical counterparts mainly capture noise. In that case, the covariance matrix tends to deliver the highest FCR and AHFA reduces to PCA. By contrast, when errors are Gaussian and factors are non-Gaussian (e.g., skewed or heavy-tailed), higher-order cumulants are driven primarily by the factor structure. Their corresponding FCR then converges to one as the sample size increases, whereas the covariance-based FCR remains bounded away from one due to the persistent contribution of idiosyncratic variance. In this case, AHFA selects the higher-order cumulant. In finite samples, the closer the factor distribution is to Gaussian, the more likely AHFA is to select PCA.

6 Simulation studies

This section reports results from several Monte Carlo simulations that assess the performance of our proposed HFA methodology in finite samples. Due to space constraints, we focus on the scenario of homogeneous factor strengths ($\alpha_1 = \cdots = \alpha_R = \alpha$) in the simulations presented in this section. Additional simulations are provided in Section 8 of the Supplementary Appendix.

6.1 Simulation setup

We use a generalization of the factor model specified in Bai and Ng (2002) and Ahn and Horenstein (2013) to generate the observations x_{it}:

$$x_{it} = \sum_{j=1}^{R}(N^{-\alpha/2}\lambda_{ij})f_{jt} + \sqrt{\theta_i}e_{it}\,, \quad e_{it} = \sqrt{\frac{1-\xi^2}{1+2J\beta^2}}u_{it}\,,$$
$$u_{it} = \xi u_{it-1} + v_{it} + \beta\sum\nolimits_{h=\max(i-J,1)}^{i-1} v_{ht} + \beta\sum\nolimits_{h=i+1}^{\min(i+J,N)} v_{ht}\,, \tag{14}$$

where $v_{it} \sim i.i.d.\,\mathcal{N}(0,1)$ and α is used to control the factor strength. The parameter ξ controls the magnitude of serial correlation of e_{it}. Cross-sectional dependence of e_{it} is governed by two parameters: β, which controls its magnitude, and J, which determines the number of correlated cross-sectional units. Each idiosyncratic error e_{it} has variance θ_i.

To model factor non-normality, we generate $f_{jt} \sim i.i.d.\,SGT(\eta_j, p_j, q_j)$, where SGT denotes the standardized skewed generalized error distribution of Theodossiou (1998). Here, (η_j, p_j, q_j) are the distributional parameters for the j-th factor: η_j controls skewness, while p_j and q_j govern kurtosis. The parameters $\{\lambda_{i,j}, \eta_j, p_j, q_j\}$ are set via a calibration procedure using estimates from the FRED-MD dataset employed in our empirical analysis. Implementation details are provided in Section 8 of the Supplementary Appendix. In addition, we set the correlation structures of the error terms as $\xi = 0.3, \beta = 0.3$, $J = 10$, and $\theta_i \backsim U[0.1, 3]$. The number of factors $R = 3$, $R_{\max} = 10$, and we consider sample sizes with $T \in \{300, 500, 1000\}$ and $N \in \{100, 200\}$.

6.2 Finite sample properties of the GER and HFA estimators

In this section, we assess the finite-sample performance of the GER and HFA estimators. In the subsequent analysis, we focus on the case of $k = 3$, namely the third-order HFA and GER, which we abbreviate as HFA3 and GER3, respectively. The analysis using $k = 4$ can be found in Section 8 of the Supplementary Appendix.

First, we evaluate the finite-sample performance of GER3 and covariance-based alternatives in selecting the number of factors: the PC3 estimator of Bai and Ng (2002), the ON estimator of Onatski (2010), and the ER and GR estimators of Ahn and Horenstein (2013). We also include a comparison with the JJR method of Jondeau et al. (2018), which is based on higher-order moments (see Section 6 of the Supplementary Appendix). Our analysis focuses on how the finite-sample performance of these estimators is influenced by the parameter α, which controls the strength of the factor model. When α is small, the factors are strong, and their influence gradually weakens as α increases. Figure 2 shows the proportion of replications in which each estimator correctly identifies the number of factors for $\alpha \in [0, 0.4]$. Each estimator is evaluated using 500 replications. The results in Figure 2 indicate that the GER3 estimator performs well in both strong and weak factor models. Its advantage is especially pronounced in the weak factor case, where all covariance-based methods perform poorly. Moreover, both the JJR and PC3 methods overestimate R even when α takes small values, which is attributable to the fact that we allow for cross-sectional dependence between errors. This finding is consistent with the simulation results reported by Ahn and Horenstein (2013).

Next, we evaluate the finite-sample properties of the PCA and HFA3 estimators for factors and factor loadings, assuming that the number of factors $R = 3$ is known. The PCA estimators are denoted by $\hat{F}^{(PCA)}$ and $\hat{\Lambda}^{(PCA)}$, while the HFA3 estimators, denoted

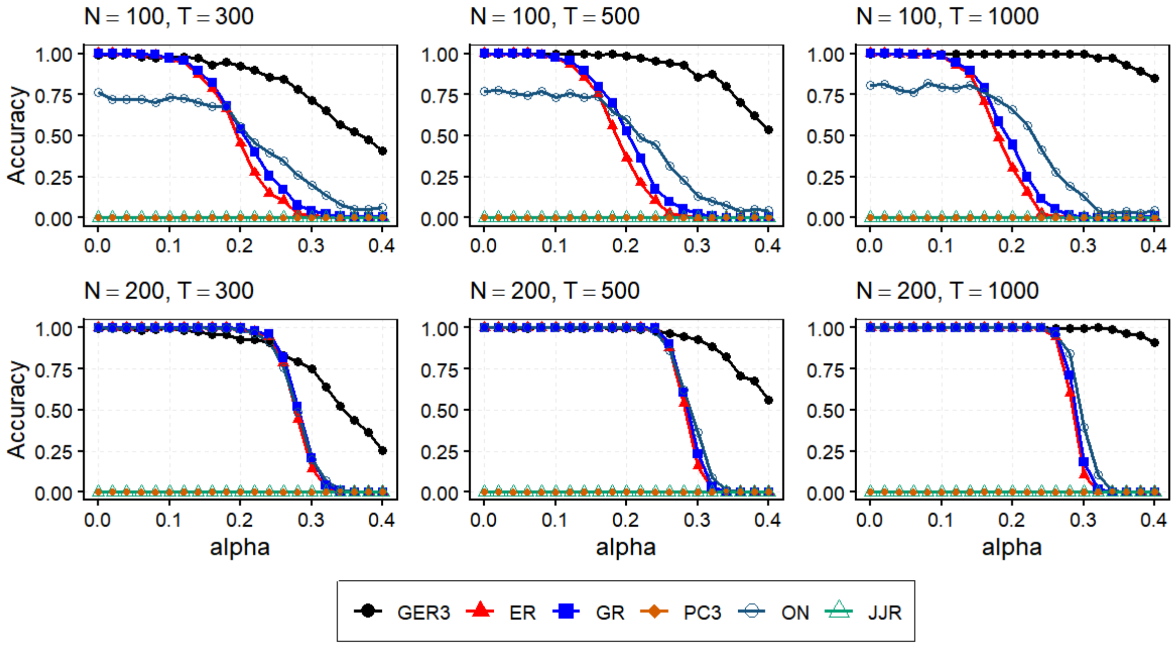

Figure 2: Proportion of replications for which the correct number of factors is estimated

Note: This figure reports the proportion of replications in which the correct number of factors is selected by six different methods: Bai and Ng (2002)'s PC3, Onatski (2010)'s ON estimator, Ahn and Horenstein (2013)'s ER and GR, Jondeau et al. (2018)'s JJR method, and the proposed GER3 estimator. The data generating process follows the factor model specified in (14). For each setting, we have 500 replications.

by $\hat{F}^{(HFA)}$ and $\hat{\Lambda}^{(HFA)}$, are obtained using third-order multi-cumulants (i.e., $k = 3$). As a measure of goodness-of-fit, we use the Trace Ratio (TR) to evaluate how closely the estimates $\hat{\Lambda}$ and $\hat{F}$ approximate their true counterparts. For example, for $\hat{F}$, the TR is defined as $\mathrm{TR}(\hat{F}, F) = \mathrm{tr}\{(F'\hat{F})(\hat{F}'\hat{F})^{-1}(\hat{F}'F)\}/\mathrm{tr}(F'F)$. This measure corresponds to a generalized squared correlation coefficient in multivariate analysis and is invariant under orthogonal rotation and scaling. It is widely used in factor analysis as a goodness-of-fit measure; see, for example, Bai and Ng (2023).

Figure 3 reports the average TR based on 500 replications for each (N, T) combination under $\alpha \in [0, 0.4]$. The main findings are as follows: First, for each sample size (N, T), the PCA estimators are less efficient than the HFA3 estimators when α is large, which is expected since the HFA3 estimators, as shown in Theorem 2, converge at a faster rate for large values of α. Second, when α is small, $\hat{\Lambda}^{(PCA)}$ tends to outperform $\hat{\Lambda}^{(HFA)}$, as $\hat{\Lambda}^{(HFA)}$ exhibits a larger asymptotic variance in such settings. Conversely, $\hat{F}^{(HFA)}$ consistently outperforms $\hat{F}^{(PCA)}$

regardless of the value of α. Third, the finite-sample properties of the estimated factors and loadings are primarily determined by T and N, respectively. Overall, the simulation results support our theoretical findings and highlight the superior performance of the HFA3 estimators, especially in weak factor scenarios.

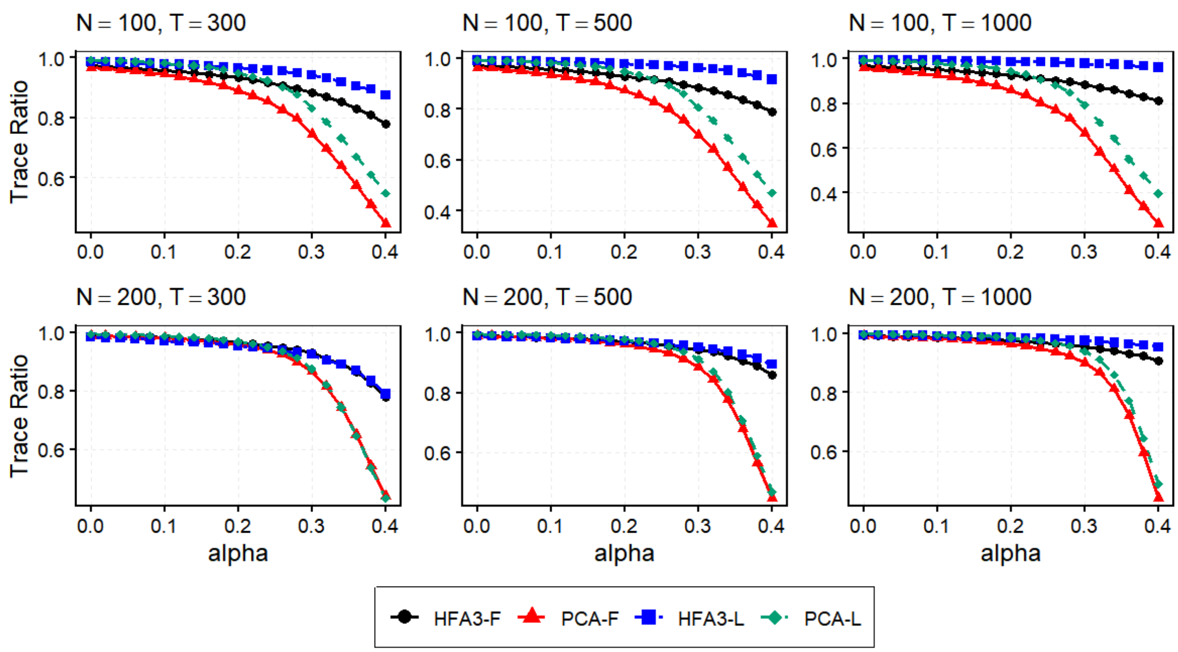

Figure 3: Accuracy of PCA and HFA3 estimators for the factors and loadings

Note: This figure reports the average Trace Ratio of factors and factor loadings by the HFA3 estimators and PCA estimators. HFA3-F (PCA-F) and HFA3-L (PCA-L), respectively, refer to the factors and factor loadings obtained via HFA3 (PCA). The data generating process follows the factor model specified in (14). For each estimation method, we have 500 replications.

6.3 Sensitivity to the degree of non-Gaussianity of the factors

In the baseline simulations, factor distributions are calibrated using the FRED-MD dataset and exhibit strong non-Gaussianity (skewness 1.461, 1.294, and -1.199), while errors are Gaussian. This setting matches the motivating scenario in which non-Gaussianity originates from the factors, under which HFA and GER improve weak-factor estimation. In practice, however, factor non-Gaussianity may be milder, making PCA preferable. The AHFA algorithm in Section 5.2 addresses this case by selecting between PCA and HFA in a data-driven way. To evaluate its performance, we conduct an additional experiment in which we gradually reduce factor non-Gaussianity and compare AHFA with PCA and HFA3.

Specifically, we continue to generate the data according to the factor model specified in (14), but allow the skewness of all factors to vary from 0 to 2; all other parameters remain fixed at the values calibrated from the FRED-MD dataset. Specifically, we consider $T \in \{300, 1000\}$, $N \in \{100, 200\}$, and $\alpha \in \{0.1, 0.2, 0.3\}$. For each given (N, T, α), we vary the SGT distribution parameter η_j from 0 to 0.98 so that the skewness of each factor increases from 0 to 2. Now for each given η_j, we compute the frequency with which the AGER, GER3 and ER estimators correctly identify the number of factors, and we report the average TR for the factors and loadings estimated by AHFA, HFA3 and PCA. Here, the AGER and AHFA estimators defined in Section 5.2 are computed with $K = 3$.

Figure 4 reports the impact of factor skewness on the probability that each estimator selects the correct number of factors. We can observe that: (i) When factor skewness is close to zero, GER3 has a low probability of identifying the correct number of factors. As factor skewness intensifies, the performance of GER3 typically improves rapidly, and it performs well in weak-factor settings; (ii) The ER estimator is unaffected by factor skewness. It performs well when the factor strength is high ($\alpha = 0.1$), but its performance deteriorates as α increases; (iii) AGER incorporates the advantages of both GER3 and ER, and exhibits robust performance across various scenarios, which confirms its reliability in empirical applications.

Next, we evaluate the impact of factor skewness on the accuracy of the estimated factors obtained via AHFA, HFA3 and PCA. Several key findings can be summarized from the results presented in Figure 5. First, when factor skewness is close to zero, the TR of HFA3 is lower than that of PCA in most scenarios, indicating lower accuracy in factor estimation. Second, as factor skewness increases, the TR of HFA3 rises significantly and exceeds that of PCA in many settings, whereas the TR of PCA is largely insensitive to skewness and remains nearly constant overall. Third, AHFA performs well across all scenarios. When factor skewness is of small magnitude, the performance of AHFA is comparable to that of PCA; when factor skewness is large, AHFA performs comparably to HFA3. Section 8 of the Supplementary Appendix reports sensitivity analyses showing that the AHFA estimator performs also well in factor loading estimation.

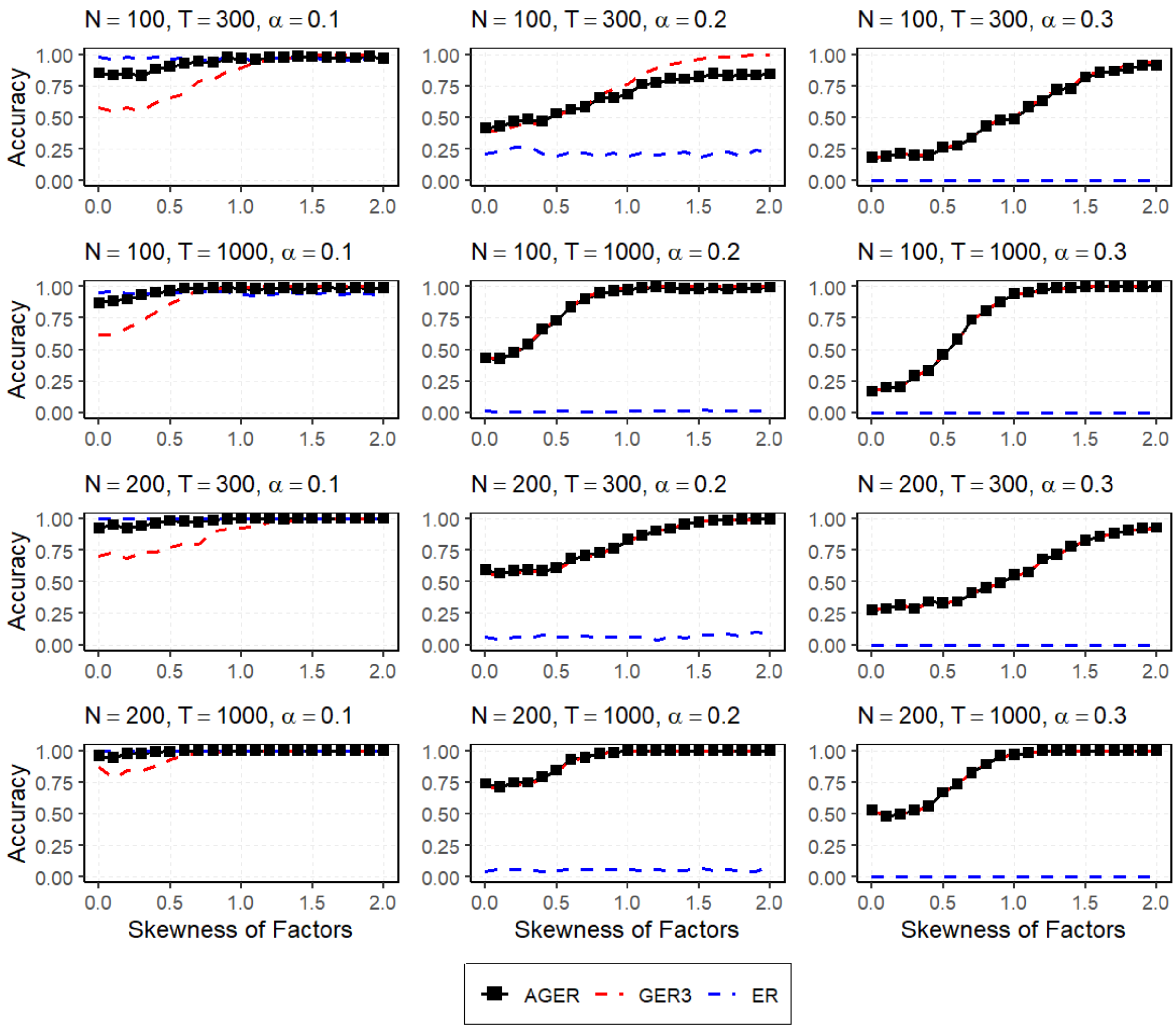

Figure 4: Impact of factor skewness on the probability that each estimator selects the correct number of factors

Note: This figure reports the proportion of replications where the correct number of factors is estimated, across different levels of factor skewness. We consider the estimators AGER, GER3 and ER. The data are generated from the factor model specified in (14). The skewness of all three factors increases from 0 to 2. For each setting, we have 500 replications.

7 Equity premium forecasting

In this section, we illustrate the usefulness of HFA for predicting the U.S. equity risk premium (ERP), defined by the excess return on the S&P 500 versus the U.S. Treasury Bill rate. It is computed as $ERP_t = \log(1 + r_t^m) - \log(1 + r_t^f)$, where r_t^m is the total return (including capital and dividend gains) on the S&P 500 portfolio — $r_t^m = (P_t - P_{t-1} + D_t)/P_{t-1}$ — where

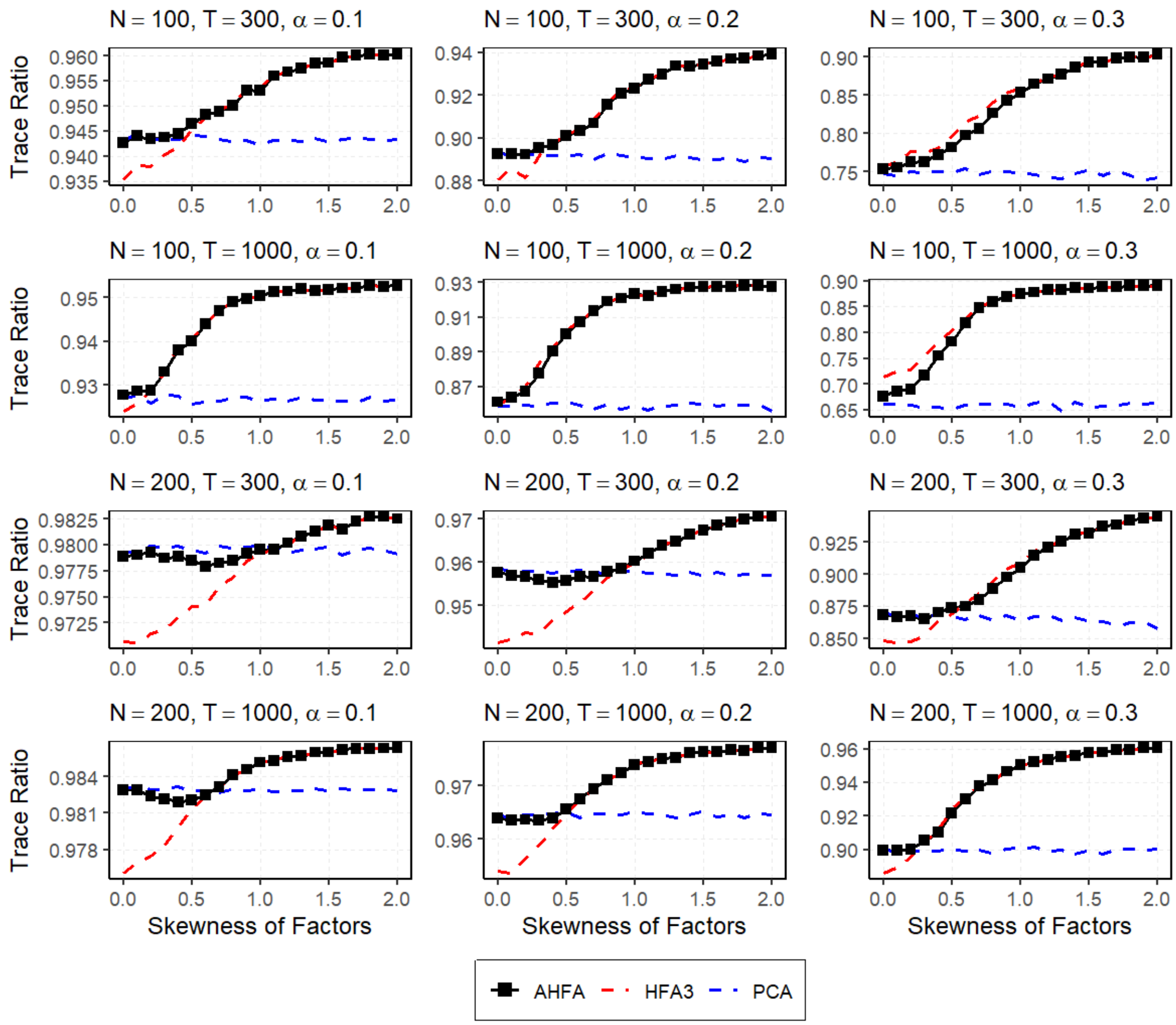

Figure 5: Impact of factor skewness on the accuracy of the estimated factors by AHFA, HFA3 and PCA

Note: This figure reports the average Trace Ratio of the estimated factors obtained via AHFA, HFA3 and PCA, across different levels of factor skewness. The data are generated from the factor model specified in (14). The skewness of all three factors increases from 0 to 2. For each setting, we have 500 replications.

P_t is the S&P 500 index value, D_t are the dividends gained during the return period, and r_t^f denotes the U.S. Treasury Bill rate.

We evaluate the predictive value of the HFA factors extracted using the FRED-MD dataset. FRED-MD is a macroeconomic database of monthly U.S. indicators. The dataset can be downloaded for free from the website http://research.stlouisfed.org/econ/mccracken/sel/. All series in FRED-MD are transformed to be stationary following the transformations described in McCracken and Ng (2016). We exclude variables that contained over 30 missing values, resulting in a FRED-MD dataset of 124 variables remaining for subsequent analysis. The series starts in January 1959 and ends in December 2018, with a total of 720 monthly observations.

We consider factor-augmented regression for forecasting. The regressions take the form

$$ERP_{t+1} = \mu + \beta(L)\hat{f}_t + \gamma(L)ERP_t + \epsilon_{t+1}\,, \tag{15}$$

where ERP_{t+1} is the 1-step-ahead variable to be forecast, $\hat{f}_t$ are the factors extracted using FRED-MD, and $\beta(L)$ and $\gamma(L)$ are finite order lag polynomials. Some researchers utilize information from the target variable to direct the dimension-reduction process towards the most valuable predictors. Prominent examples of such methods are Partial Least Squares (Kelly and Pruitt, 2015) and Supervised PCA (Giglio et al., 2025). These approaches leverage the target-related information to optimize the dimension-reduction procedure, ensuring that the resulting components are more relevant for prediction.

All factor-augmented regression models in (15) are estimated through a rolling-window estimation approach starting in January 1985 and comprising a 408-month out-of-sample period. We consider two out-of-sample periods. The first out-of-sample prediction begins in January 1985 and ends in October 2007 (Pre-crisis period). The second out-of-sample period starts in November 2007 and ends in December 2018, intended to evaluate predictability during the crisis and recovery periods.

We perform several diagnostic checks on the FRED-MD dataset using the estimated factors and residuals obtained from the HFA method. First, Figure 6 provides strong evidence for the assumptions of non-Gaussian factors. Second, we formally test each residual series for normality using the test proposed by Bai and Ng (2005) at the 1% significance level, and report the number of rejections. After extracting three factors using HFA3 and HFA4, the normality hypothesis is rejected for only 10 and 8 out of the 124 residuals, respectively. Taken together, these graphical and statistical checks suggest that the key assumptions underpinning the validity of the HFA methodology are reasonably well satisfied. Further analysis of the FRED-MD dataset is provided in Section 9 of the Supplementary Appendix.

We further compare the predictive content of three factor structures: the first, second, and

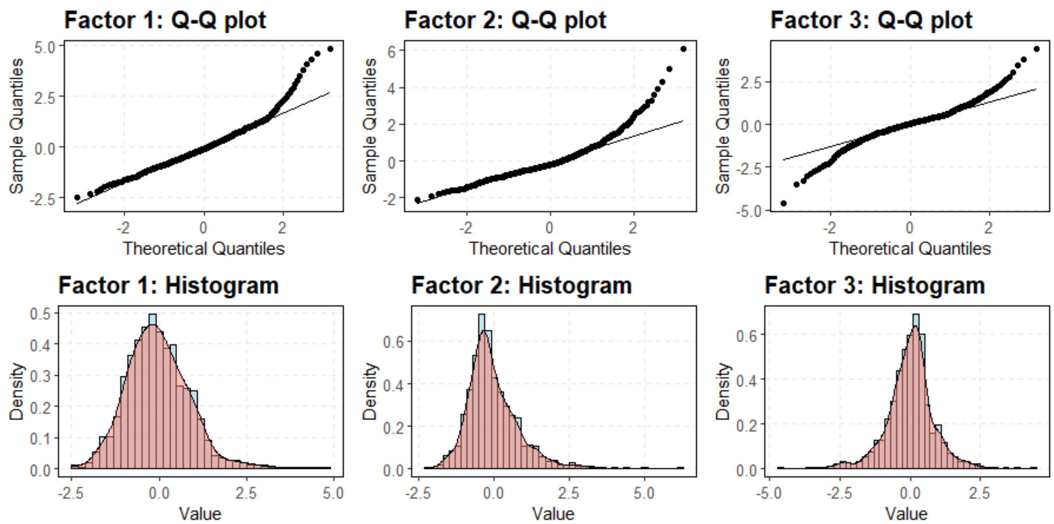

Figure 6: The Q-Q plots and histograms of the HFA3 factors from the FRED-MD dataset

Table 1: Out-of-sample MSE for equity premium forecasting

	Pre-Crisis		Crisis & Post-Crisis		Full Sample	
	MSE	OS R^2	MSE	OS R^2	MSE	OS R^2
Panel A: AGER criterion factor structure						
AHFA	10.666	0.061	14.659*	0.044	11.978*	0.054
HFA3	10.669	0.060	14.958*	0.024	12.078*	0.046
HFA4	**10.639**	**0.063**	**14.661***	**0.044**	**11.960***	**0.055**
PCA	10.826	0.047	15.428	-0.006	12.338	0.025
JMCA	10.749	0.053	15.430	-0.007	12.286	0.030
Panel B: ER criterion factor structure						
AHFA	10.711	0.057	14.819*	0.033	12.060*	0.047
HFA3	10.669	0.060	15.027	0.020	12.100*	0.044
HFA4	10.639	0.063	14.734*	0.039	11.984*	0.053
PCA	10.790	0.050	15.355	-0.002	12.289	0.029
JMCA	10.788	0.050	15.357	-0.002	12.288	0.029
Panel C: JJR criterion factor structure						
AHFA	10.941	0.036	15.249	0.005	12.356	0.024
HFA3	10.863	0.043	14.948	0.025	12.204	0.036
HFA4	10.927	0.038	15.249	0.005	12.347	0.025
PCA	10.992	0.032	15.018	0.020	12.314	0.027
JMCA	10.909	0.039	14.984	0.023	12.247	0.033

Note: This table reports the MSE and Out-of-Sample R^2 of five alternative 1-month-ahead forecasting methods for the equity premium (the best-performing methods are shown in bold). The out-of-sample forecasting is implemented using rolling windows of 310 observations. The MSE ratios with an asterisk indicate that the left-sided DM test is significant at the 10% level, with PCA serving as the benchmark model. Panels A, B, and C use the number of factors estimated by the AGER, ER, and JJR estimators, respectively.

third ones are based on the AGER, ER, and JJR criteria, respectively. For each factor structure, we consider AHFA, HFA3, HFA4, JMCA, and PCA to extract the factors and compare their predictability. Here the AGER and AHFA estimators defined in Section 5.2 are computed with $K = 4$. For each month, we re-estimate the number of factors using the proposed AGER estimator, Ahn and Horenstein (2013)'s ER estimator, and Jondeau et al. (2018)'s JJR approach. We find that the ER estimator always detects one factor that has strong influential power. The AGER estimator detects between one and four factors, typically including several weak factors. The JJR approach always selects four factors. Based on these results, we re-estimate the factors each month. We use the predictive mean square error (MSE) and out-of-sample R-squared coefficient (R^2_{OOS}) to evaluate the prediction performance. The R^2_{OOS} is defined as $R^2_{OOS} = 100 \times (1 - \sum_{t=1}^{T^*}(\widehat{ERP}_t - ERP_t)^2 / \sum_{t=1}^{T^*}(\widehat{ERP}_t^{HA} - ERP_t)^2)$, where T^* is the number of out-of-sample forecasts, and $\widehat{ERP}_t^{HA} = \frac{1}{t-1}\sum_{s=1}^{t-1} ERP_s$ is the historical average forecast. To avoid bias, we determine the regression structure after the factor number is selected. To that end, we use BIC to select the number of autoregressive lags ($1 \leqslant p \leqslant 6$) and lags of the factors ($1 \leqslant m \leqslant 3$) for the rolling-window sample. Further, to compare the predictive difference between PCA and HFA, we use the t-type test statistic of Diebold and Mariano (1995) to determine whether the forecast differences are statistically significant.

The results assessing differences in predictive power are reported in Table 1. The MSE ratio marked with an asterisk indicates that the DM test is significant at the 10% level, implying that the predictive performance of this method is superior to that of PCA. As can be observed, regardless of whether we consider Panels A, B, or C, the regression models based on HFA factors achieve the best rank compared to PCA for the three periods under consideration. The predictive difference between PCA and HFA (AHFA) can also be observed from the DM tests in Panel A and Panel B. Additionally, comparing Panel A with Panels B and C, it is clear that the forecast models with the AGER criterion factor structure outperform the ER and the JJR criterion, especially in the crisis and post-crisis periods.

8 Conclusion

This study develops a new framework for factor analysis based on the eigenvalue decomposition of the product of the higher-order multi-cumulant and its transpose. The proposed Generalized Eigenvalue Ratio (GER) and Higher-order multi-cumulant Factor Analysis (HFA) methodology consistently estimate the number of factors, the factors themselves, and factor loadings for high-dimensional non-Gaussian panel data with an underlying weak factor structure. The convergence rates and asymptotic distributions of the estimated factors and factor loadings are derived under the assumption of Gaussian idiosyncratic errors. HFA is recommended when non-normality is driven by weak latent factors. To guide this choice, we propose a data-driven procedure, Adaptive HFA, that determines when HFA should be preferred over Principal Component Analysis (PCA). All results are validated through extensive simulation studies and an empirical application. An interesting direction for future research is to exploit higher-order cumulant information for factor analysis under nonlinear factor models.

Acknowledgements

Guanglin Huang's research is sponsored by the National Social Science Fund of China (25FYB023), the Natural Science Foundation of Sichuan Province (2026NSFSC0791) and the China Postdoctoral Science Foundation (2023M742878). Wanbo Lu's research is sponsored by the National Science Foundation of China (71771187, 72011530149) and the Fundamental Research Funds for the Central Universities (JBK190602) in China. Kris Boudt received funding from the Research Foundation Flanders (G0G8320N). We are grateful to David Ardia, Dries Cornilly and Serena Ng for helpful comments on a previous draft. We also thank participants at the econometrics seminar of Vrije Universiteit Amsterdam, the EC2 2020 conference at HEC Paris, and the HDTS 2021 workshop at Maastricht University.

References

Ahn, S. C. and Horenstein, A. R. (2013). Eigenvalue ratio test for the number of factors. *Econometrica*, 81, 1203–1227.

Bai, J. (2003). Inferential theory for factor models of large dimensions. *Econometrica*, 71, 135–171.

Bai, J. and Ng, S. (2002). Determining the number of factors in approximate factor models. *Econometrica*, 70, 191–221.

Bai, J. and Ng, S. (2005). Tests for skewness, kurtosis, and normality for time series data. *Journal of Business and Economic Statistics*, 23, 49–60.

Bai, J. and Ng, S. (2013). Principal components estimation and identification of static factors. *Journal of Econometrics*, 176, 18–29.

Bai, J. and Ng, S. (2023). Approximate factor models with weaker loadings. *Journal of Econometrics*, 235, 1893–1916.

Bailey, N., Kapetanios, G., and Pesaran, M. H. (2021). Measurement of factor strength: Theory and practice. *Journal of Applied Econometrics*, 36, 587–613.

Barigozzi, M. and Massacci, D. (2025). Modelling large dimensional datasets with Markov switching factor models. *Journal of Econometrics*, 247, 105919.

Barone Adesi, G., Gagliardini, P., and Urga, G. (2004). Testing asset pricing models with coskewness. *Journal of Business and Economic Statistics*, 22, 474–485.

Boudt, K., Cornilly, D., and Verdonck, T. (2020). Nearest comoment estimation with unobserved factors. *Journal of Econometrics*, 217, 381 – 397.

Chang, J., Guo, B., and Yao, Q. (2015). High dimensional stochastic regression with latent factors, endogeneity and nonlinearity. *Journal of Econometrics*, 189, 297–312.

Chang, J., Guo, B., and Yao, Q. (2018). Principal component analysis for second-order stationary vector time series. *Annals of Statistics*, 46, 2094–2124.

Chen, L., Dolado, J. J., and Gonzalo, J. (2014). Detecting big structural breaks in large factor models. *Journal of Econometrics*, 180, 30–48.

Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business and Economic Statistics*, 13, 253–263.

Fan, J., Guo, J., and Zheng, S. (2022). Estimating number of factors by adjusted eigenvalues thresholding. *Journal of the American Statistical Association*, 117, 852–861.

Fortin, A.-P., Gagliardini, P., and Scaillet, O. (2025). Eigenvalue tests for the number of latent factors in short panels. *Journal of Financial Econometrics*, 23, nbad024.

Freyaldenhoven, S. (2022). Factor models with local factors—Determining the number of relevant factors. *Journal of Econometrics*, 229, 80–102.

Giglio, S., Xiu, D., and Zhang, D. (2025). Test assets and weak factors. *Journal of Finance*, 80, 259–319.

Giovannelli, A., Massacci, D., and Soccorsi, S. (2021). Forecasting stock returns with large dimensional factor models. *Journal of Empirical Finance*, 63, 252–269.

Granger, C. W. (1976). Tendency towards normality of linear combinations of random variables. *Metrika*, 23, 237–248.

Hyvärinen, A. (1999). Fast and robust fixed-point algorithms for independent component analysis. *IEEE Transactions on Neural Networks*, 10, 626–634.

Hyvärinen, A., Zhang, K., Shimizu, S., and Hoyer, P. O. (2010). Estimation of a structural vector autoregression model using non-gaussianity. *Journal of Machine Learning Research*, 11, 1709–1731.

Jondeau, E., Jurczenko, E., and Rockinger, M. (2018). Moment component analysis: An illustration with international stock markets. *Journal of Business and Economic Statistics*, 36, 576–598.

Kelly, B. and Pruitt, S. (2015). The three-pass regression filter: A new approach to forecasting using many predictors. *Journal of Econometrics*, 186, 294–316.

Kolda, T. G. and Bader, B. W. (2009). Tensor decompositions and applications. *SIAM Review*, 51, 455–500.

Lettau, M. and Pelger, M. (2020). Estimating latent asset-pricing factors. *Journal of Econometrics*, 218, 1–31.

Li, Z., Ton, J.-F., Oglic, D., and Sejdinovic, D. (2021). Towards a unified analysis of random fourier features. *Journal of Machine Learning Research*, 22, 4887–4937.

Liu, X. and Chen, R. (2020). Threshold factor models for high-dimensional time series. *Journal of Econometrics*, 216, 53–70.

Ludvigson, S. C. and Ng, S. (2007). The empirical risk–return relation: A factor analysis approach. *Journal of Financial Economics*, 83, 171–222.

Ma, C. and Tu, Y. (2023). Group fused Lasso for large factor models with multiple structural breaks. *Journal of Econometrics*, 233, 132–154.

McCracken, M. W. and Ng, S. (2016). FRED-MD: A monthly database for macroeconomic research. *Journal of Business and Economic Statistics*, 34, 574–589.

Mooijaart, A. (1985). Factor analysis for non-normal variables. *Psychometrika*, 50, 323–342.

Onatski, A. (2010). Determining the number of factors from empirical distribution of eigenvalues. *Review of Economics and Statistics*, 92, 1004–1016.

Onatski, A. (2012). Asymptotics of the principal components estimator of large factor models with weakly influential factors. *Journal of Econometrics*, 168, 244–258.

Risk, B. B., Matteson, D. S., and Ruppert, D. (2019). Linear non-gaussian component analysis via maximum likelihood. *Journal of the American Statistical Association*, 114, 332–343.

Su, L., Shi, Z., and Phillips, P. C. (2016). Identifying latent structures in panel data. *Econometrica*, 84, 2215–2264.

Theodossiou, P. (1998). Financial data and the skewed generalized t distribution. *Management Science*, 44, 1650–1661.

Uematsu, Y. and Yamagata, T. (2022). Estimation of sparsity-induced weak factor models. *Journal of Business and Economic Statistics*, 41, 213–227.

Wall, M. M. and Amemiya, Y. (2000). Estimation for polynomial structural equation models. *Journal of the American Statistical Association*, 95, 929–940.

Zhang, B., Pan, G., Yao, Q., and Zhou, W. (2024). Factor modeling for clustering high-dimensional time series. *Journal of the American Statistical Association*, 119, 1252–1263.